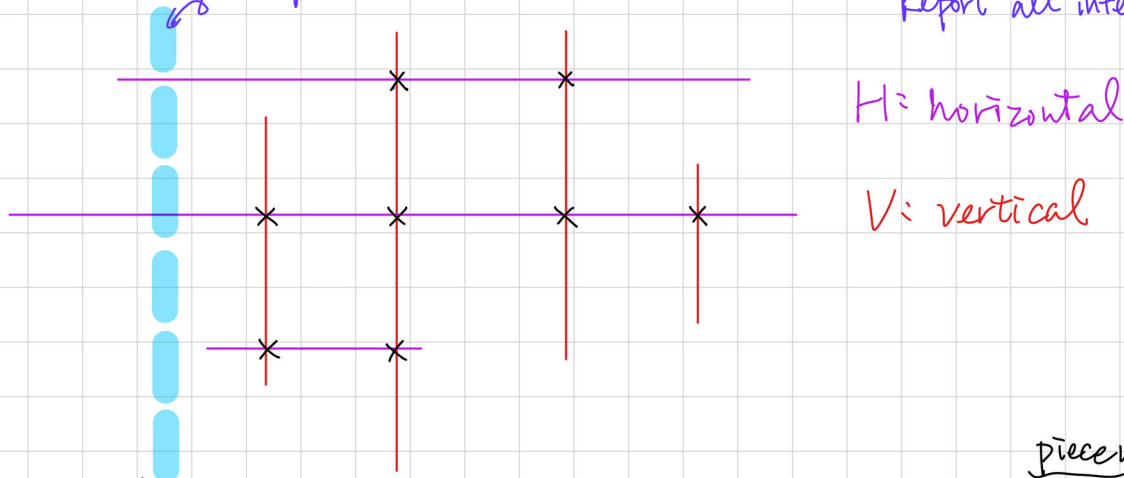


Lecture 1 - (Planesweep) Line Segment Intersection (I)

Simplification: Orthogonal Segment Intersection.
Report all intersection pts.



Note: Intersection points may reach $\Theta(n^2)$. [Consider $\binom{n}{2}$].

Question: How can we design a better method to determine the complexity?

By output size k .

→ Design an algorithm to achieve the time complexity $\mathcal{O}(n \log n + k)$

Invariant. $H(l)$: The set of segments (type H) intersecting with the sweep line l .

Output occurs when l reaches a type V segment. Let this segment be $v = x \times [y_1, y_2]$. Report all $H(l) \cap [y_1, y_2]$.

Operations: a data structure D (1-dimensional) that supports

- Insertion & Deletion of segments

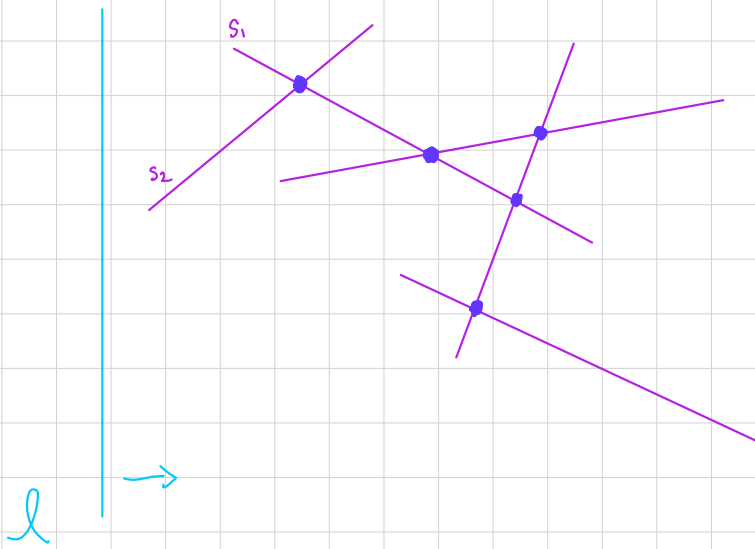
- Reporting the segments lying within a given range

→ BST! Sort all events in advance.

Let $n = |H| + |V|$
 $\mathcal{O}(n \log n + n \log |H| + k)$

Lecture 2 - (Planesweep) Line Segment Intersection (II)

Original Version: Arbitrary Line Segment Intersection.



Question: What kind of information to store?

How to update the respective information?

- Relevant events

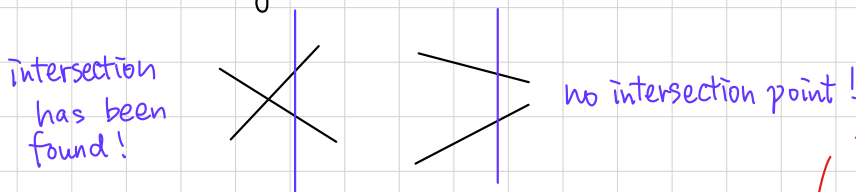
→ Design an algorithm to achieve time complexity $O(n \log n + k \log n)$

① Claim: There is a moment where s_1, s_2 are consecutive when swept by l .

② Invariant: Relative ordering of the segments w.r.t. the sweepline.

Σ : current ordering (Sweepline status)

For every pair of consecutive segments in Σ , σ_1 and σ_2 , we are interested in their intersection point in the future, if any.



Maintain the sweepline status Σ and event queue Q .

Events: - Left endpoint of l .

$O(n)$ Insert l to Σ
Detect possible intersection immediately above/below l and add to Q if possible

- Right endpoint of l .

$O(n)$ Remove (l) in Σ
Detect possible intersection immediately above/below l and add to Q if possible

Everytime when Σ changes check if new instances of Q can be created!

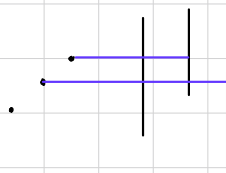
- Intersection points
Swap segment

$O(k)$

Maintaining BBST: Insert/Delete/Swap/Predecessor/Successor

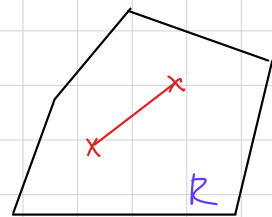
Σ : key storage $\rightarrow$ line equation instead of coordinates

Q : Report the element with minimum x-coordinate in Q



Lecture 3 - (Planesweep) Convex Hull

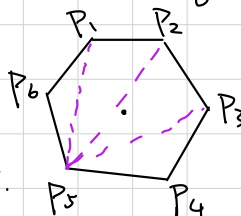
Convexity: Any 2 points $x, y \in R$ and segment $\overline{xy}$ are entirely contained in R .



Convex Hull: The smallest convex polygon covering P .

Prove any $p \in R$ where R is bounded by H that

$$\vec{p} = \sum_{i=1}^{|H|} \vec{h}_i w_i \text{ with } w_i \geq 0, \sum w_i = 1.$$

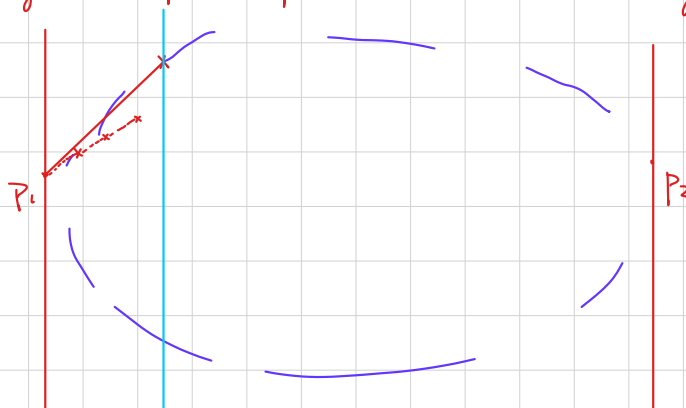


Constrained

Generality.

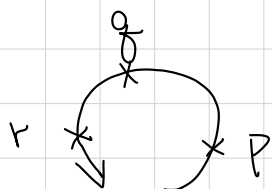
Just need to consider upper-hull

Only one leftmost point. (Resolve the issue by rotating small θ).



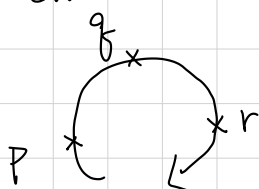
Requirement for planesweep: Sort x -coordinate.
 $O(n \log n)$

CCW



$$\text{orient}(p, q, r) > 0$$

CW

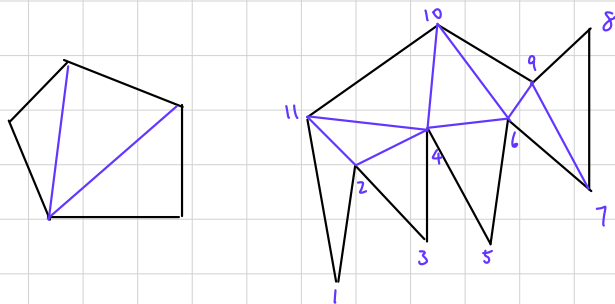


$$\text{orient}(p, q, r) < 0$$

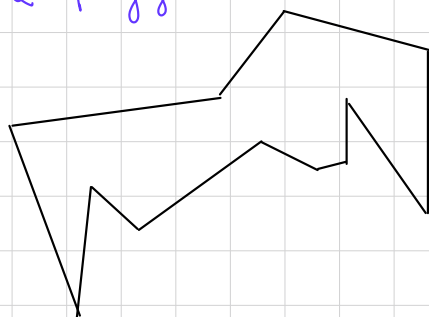
Monotonic Stack.

Lecture 4 - (Planesweep) Polygon Triangulation

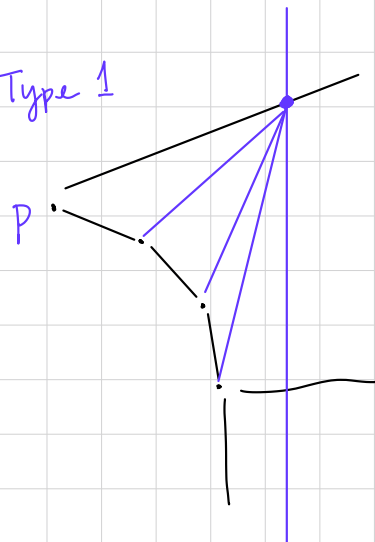
? Prove. You can always find one diagonal (ear)



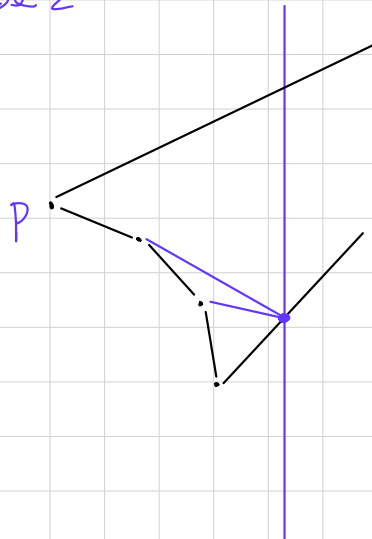
x-monotone polygon



Type 1



Type 2



Note: A simple x-monotone polygon's points can be sorted in $O(n)$ time.

By merging two chains.

(Given CW/CCW sequence) $\mathcal{L}$



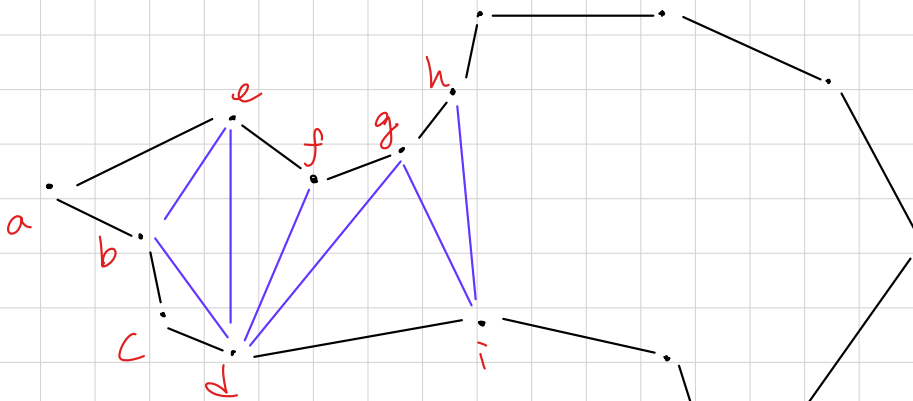
$U \cup \mathcal{L}^{rev}$

Question: How to maintain the "horns"?

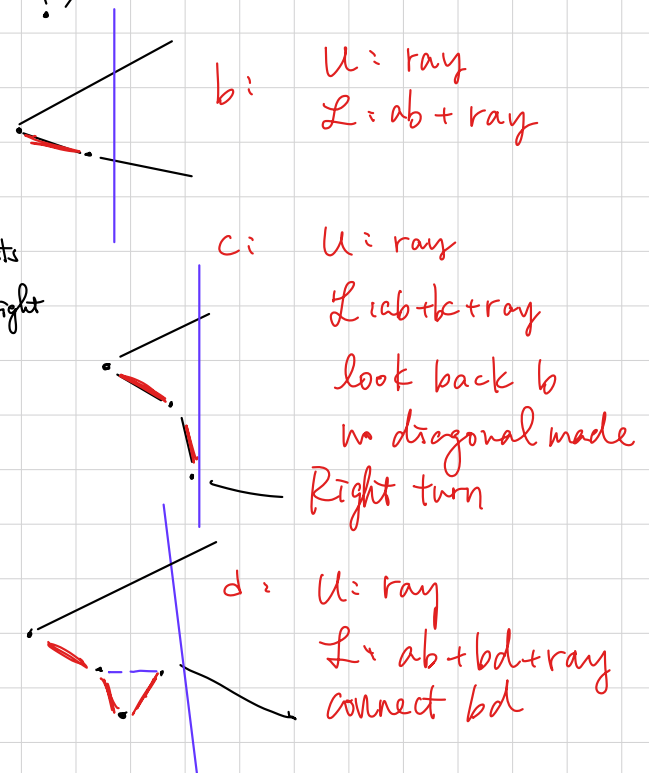
Case 1: Next point encountered is situated at the upper boundary.

Case 2: Next point encountered is situated

Lecture 5 - (Planesweep) Polygon Triangulation



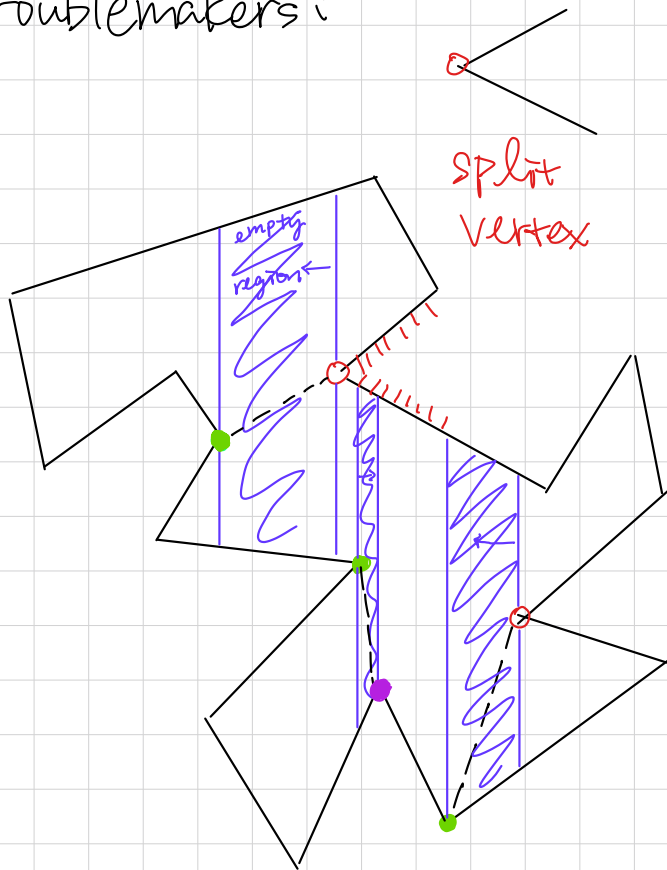
Invariant: Untriangulated region lying to the vertex just processed, v_i , consists of two x-monotone chains, a lower chain & upper chain each containing at least one edge. If the chain from v_i to u has two or more edges, then these edges form a reflex chain. The other chain consists of a single edge whose left point is u and whose right endpoint lies to the right of v_i .



Time complexity: $O(1 + \# \text{ diagonal} + \text{preprocessing})$
 $= O(k)$

? After solving one case?

Troublemakers:



split
vertex

merge
vertex

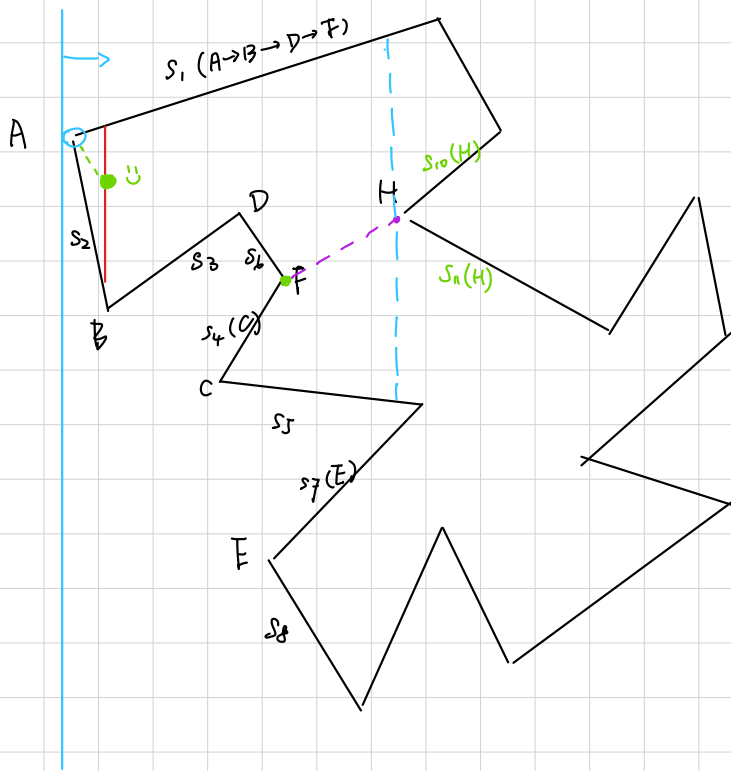
Remove split vertex

Claim: all split vertices have
been removed.
all merge vertices have
been removed.

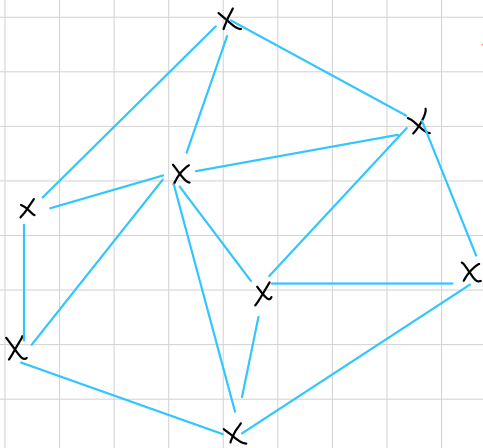
- Each edge belongs to at most 2 sub-polygons. $O(n \log n)$.



Maintain a point helper always visible to the split vertex
getting hit by the sweep line



- Identify segments & endpoints
- BBT query segment above $q \in \mathbb{R}^2$.
update helper(e) (tightest visible vertex in between)



Exercise.

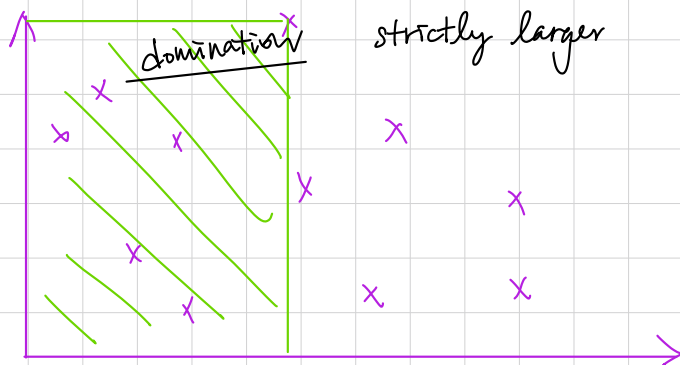
Triangulation of a point set.
in $O(n \log n)$ time.

Hint: Define a new helper.

Lecture 6 - (Dimensionality Reduction) Dominance Screening

- Low dimensionality: constant dimension
- High dimensionality: exponentially dependent on dimension

Maxima Problem



Dominance Screening

P, Q point set.

Report all $q \in Q$ st. q is not dominated by any $p \in P$.

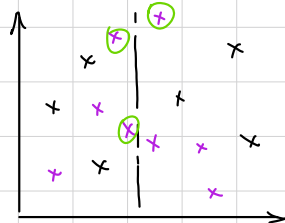
1D: Solved in $O(n)$ Linear Scan

① Scan all $p \in P$ and calculate $M = \max_p x_p$.

② Report all $q \in Q$ with $M < x_q$.



2D: D&C Evenly spread $P \cup Q$.



Obs 1. P_R is the final answer

Obs 2. Eliminate P_L by white points on R

$2D \rightarrow 1D$.



project P_L on l
project W_R on l .

Analysis. $T(n) = 2T(\frac{n}{2}) + n \Rightarrow T(n) = O(n \log n)$

3D: Analysis $T(n) = 2T(\frac{n}{2}) + O(n \log n)$

$\Rightarrow T(n) = O(n \log^2 n)$

* : Planesweep $O(n \log n)$

Sort x & y & z
Use SMT.

Lecture 7 - (Dimensionality Reduction) Rectangle Point Containment

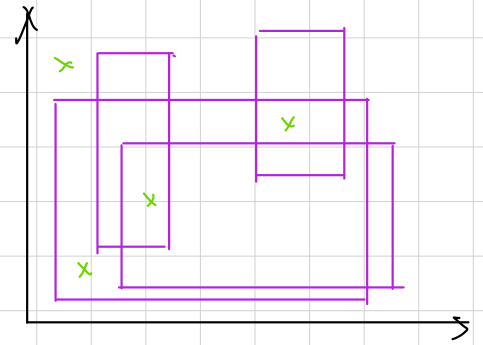
Rectangle Point Containment Problem

$$r \in \mathbb{R}^d$$

$$r = [x_1, x_2] \times [y_1, y_2] \times [z_1, z_2] \times \dots \in \mathbb{R}^d$$

$$p \in P$$

Report all (p, r) where $p \in r$.



Solve the problem in $O(n \text{ polylog } n + \Theta(\text{output sensitivity}))$ time.

n - x -coord size

Assumption: No horizontal stripes.*

D&C

* Note that the sub-problem might not be defined!
Remove these unbounded rectangles by dimensionality reduction.

1D



Lecture 8- Output Sensitive Algorithms (Maxima)

$$P = \{(x, y) \in \mathbb{R}^2\}$$

$$n = |P|$$

p dominates q .

$$\text{if } x_p > x_q \text{ or } y_p > y_q$$

(strictly better in at least one dimension)

Report any $p \in P$ NOT dominated by any $p' \in P$ and $p \neq p'$.

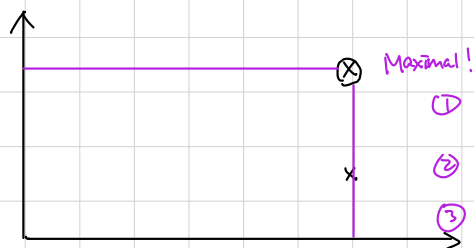
Easy: $O(n \log n)$ algorithm

- Sort by x -coord (descending)

- Remember largest k .

Goal: $O(n \log k)$ algorithms (!)

- $O(nk)$ algorithm.



① $\arg\max_p x_p$

$O(n)$

② Remove all pts dominated by p

$O(n)$

③ Repeat.

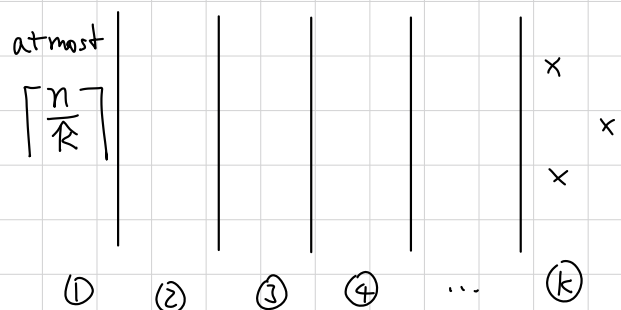
$O(nk)$

- $\hat{k}$: an upper bound of k .

$$O(n \log \hat{k}).$$

★ Multi-rank Selection.

How?



- At most $\lceil \frac{n}{k} \rceil$ points in group k .

Eliminate group $k-1$ by identifying the highest point in group k and apply the $O(nk)$ algorithm

- Analysis

Overall running time

① $O(n \log \hat{k})$ partitioning

$$\textcircled{2} \sum_{i=1}^{\hat{k}} \frac{n}{\hat{k}} \cdot s_i = O(n).$$

- Guessing the $\hat{k}$.

Exercise: Given a guess s , verify whether $s \geq k$ in $O(n \log s)$.

If $s \geq k$, terminate

Otherwise, increase s .

Fact. $1+2+4+\dots+2^{\log k+1} = O(k)$

1. $\sum_{i=1}^{\log k} \frac{n}{2^i}$

$$O(n) + O(n \lg 2) + O(n \lg 4) + O(n \lg 8) + \dots$$
$$O(n \log^2 n)$$

2. Observation

Let non-linear part of the algorithm form a GP.
eliminating the log.

$s=1$	$O(n)$
$s=2$	$O(n \lg 2)$
$s=2^2$	$O(n \lg 2)$
$\vdots$	
$s=2^i$	$O(2^i n \lg 2)$

Observation: $2^{2^{i-1}} < k$.

$$n(1+2+\dots+2^{2^i}) = O(2^i \cdot n) = O(n \log k)$$

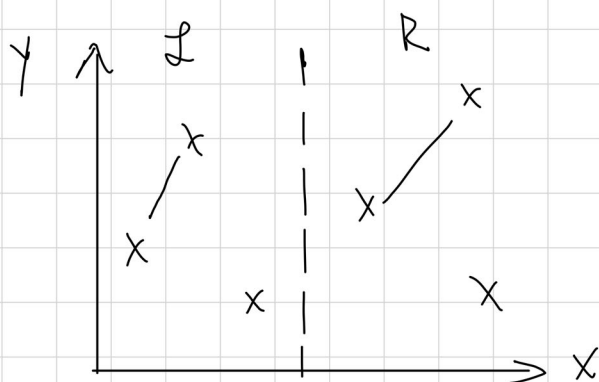
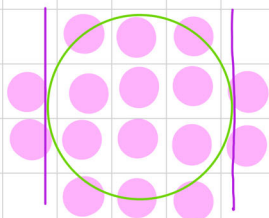
Lecture 9 - (Grid Decomposition) Closest Pair

Packing Lemma: Impose a regular grid on $\mathbb{R}^d$ where every cell is a box with side length s on each dimension.

Any ball with radius r can overlap with no more than

$$\left(1 + \left\lceil \frac{2r}{s} \right\rceil\right)^d \text{ cells.}$$

Proof. * (Pictorially)



2D. Divide P evenly using a vertical line l into P_1, P_2 . Recursively find closest pairs in P_1, P_2 .

Consider the cross pairs $P_1 \times P_2$, within distance of r .

→ Impose a grid G (i) each cell is an axis-parallel square (ii) side length $r/\sqrt{2}$ (iii) l is a line in the grid

Obs. 1. Each point can be covered by at most 4 cells.

Obs. 2. Every cell c contains at most 2 points.

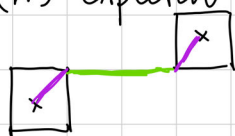
Obs. 3. At most 4n non-empty cells

Obs. 4. A cell can have at most $O(1)$ r-neighboring cells.

Hashing → $O(n)$ expected time.

for every non-empty cell c_1 on the left of l
for every r-neighbor cell c_2 of c_1 on the right of l
calculate every pt $(p_1, p_2) \in (c_1(P), c_2(P))$
return pair (p_1, p_2) with closest distance.

Merge = $O(n)$ expected



$$f(n) = 2f\left(\frac{n}{2}\right) + O(n) \Rightarrow f(n) = O(n \log n)$$

Lecture 10 - (Grid Decomposition) Well-separated Pair Decomposition Applications

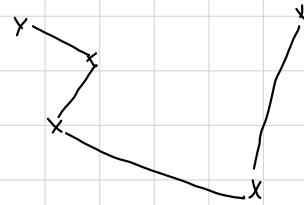
• Diameter $\text{diam}(P)$

$$\text{diam}(P) \leq x \leq \text{diam}(P)(1+\epsilon)$$

• Euclidean Spanning Tree

P : a set of n points in $\mathbb{R}^d$

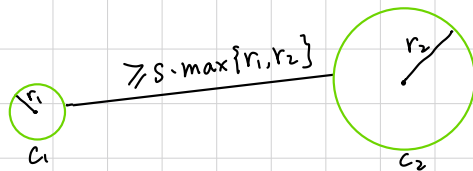
$$\text{cost}(\langle p_i, p_j \rangle) = \text{dist}(p_i, p_j)$$



Definition (well-separated): Let A, B be sets of points in $\mathbb{R}^d$.

A, B are said to be s -well separated if

$\exists C_1, C_2$, where C_1 covers all points in A with radius r_1 , C_2 covers all points in B with radius r_2
s.t. the distance of $C_1, C_2 \geq s \cdot \max\{r_1, r_2\}$



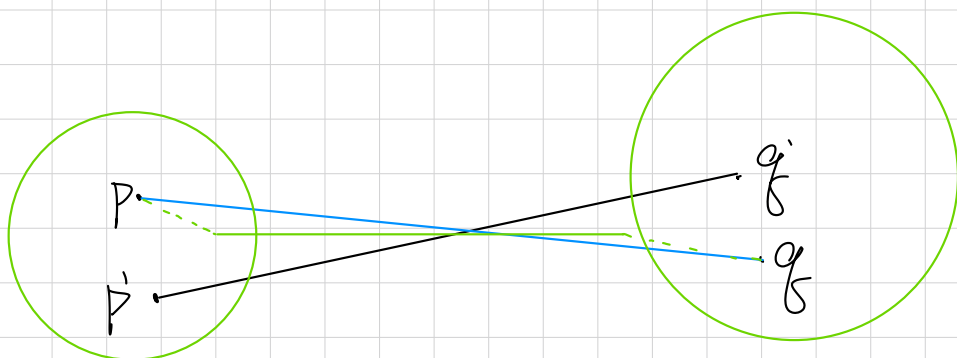
Utility Lemma: $p, p' \in A, q \in B$

$$\text{dist}(p, p') \leq 2/s \text{ dist}(p, q)$$



$$p, p' \in A, q, q' \in B$$

$$\frac{s}{s+4} \text{dist}(p', q') \leq \text{dist}(p, q) \leq \frac{s+4}{s} \text{dist}(p', q')$$



Proof. $\exists A, B$ with radius r_A, r_B .

$$\text{dist}(p', q') \geq \text{dist}(C_1, C_2).$$

$$\text{dist}(p, q) \leq \text{dist}(C_1, C_2) + 4 \cdot \max\{r_1, r_2\}$$

$$\Rightarrow \frac{\text{dist}(p, q)}{\text{dist}(p', q')} \leq \frac{\text{dist}(C_1, C_2)}{\text{dist}(C_1, C_2) + 4 \max\{r_1, r_2\}} = \frac{1}{1 + 4 \frac{\max\{r_1, r_2\}}{\text{dist}(C_1, C_2)}}$$

$$\text{dist}(C_1, C_2) \geq s \max\{r_1, r_2\}.$$

Giving $\boxed{\frac{s}{s+4}}$

Lecture 11 - (Grid Decomposition) Well-separated Pair Decomposition

Definition (s-well-separated pair decomposition)

P : a set of n points in $\mathbb{R}^d$.

$\{A_1, B_1\}, \{A_2, B_2\}, \dots, \{A_m, B_m\}$ forms a WSPD

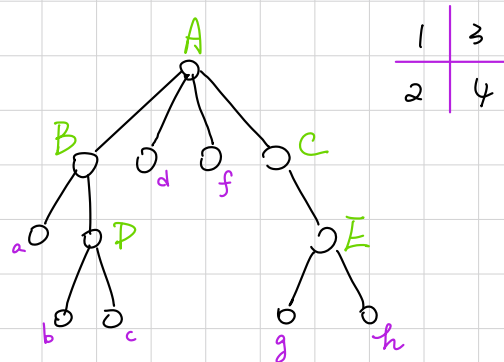
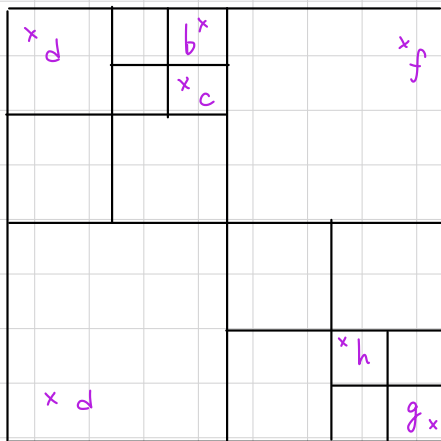
- ① $A_i \subseteq P, B_i \subseteq P, A_i \cap B_i = \emptyset$ for $i \in [m]$.
- ② A_i and B_i are well separated for $i \in [m]$
- ③ For any distinct points $p, q \in P$. $\exists i \in [m]$ s.t. $p \in A_i, q \in B_i$.

1. $m = \Theta(n^2) \quad \bigcup_{i=1}^n \bigcup_{j=i+1}^n \{\{p_i, p_j\}\}.$

2. $m = O(n \log \Delta).$

Let $d_{\max} = \max_{p, q \in P} \text{dist}(p, q)$. $d_{\min} = \min_{\text{distinct } p, q \in P} \text{dist}(p, q)$
 $\Delta = \frac{d_{\max}}{d_{\min}}$

* Prove that construction of quad-tree can be done in $O(n \log \Delta)$ time.
 Quad-Tree



height of the tree = $O(\log \Delta)$

nodes on each lv = $O(n)$

At least 2 nodes per split.

Total time = $O(n \log \Delta)$

Lecture 12 - (Grid Decomposition) Well-separated Pair Decomposition

ws-pairs $(u, v, s) \{$

if $(u$ and v are leaves and $u=v$) return;

if $(\text{rep}(u)$ or $\text{rep}(v)$ is empty) return $\emptyset$;

else if $(u$ and v are s -well separated) return $\{u, v\}$;

else {

if $(\text{level}(u) > \text{level}(v))$ swap (u, v) ;

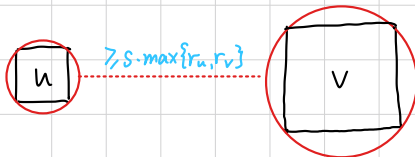
Let $u_1, u_2, \dots, u_k$ denote the children of u ;

return $\bigcup_{i=1}^k \text{ws-pairs}(u_i, v, s)$;

}

}

Two cells are s -well-separated



$$\text{dist}(C_1, C_2) \geq s \cdot \max\{r_1, r_2\}.$$

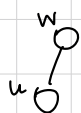
Charging Argument. $\rightarrow$ Charge the nodes that have been split.

Consider the sequence of finding the WSPD. $\{A, A\} \xrightarrow{A} \{A, C\} \xrightarrow{A} \{B, C\} \xrightarrow{B} \{a, C\} \xrightarrow{C} \{a, E\}$.

Key fact: $\text{level}(u), \text{level}(v)$ differs at most 1.

$\rightarrow$ A node can at most be charged $O(1)$ time.

Proof.

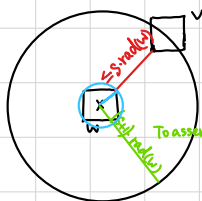


$$\{w, v\} \Rightarrow \{u, v\}.$$

$$\text{level}(w) = \text{level}(v) \text{ or } \text{level}(w) = \text{level}(v) - 1$$

$$\Rightarrow \begin{cases} \text{(i) side length of } w = \text{side length of } v, \\ \text{(ii) side length of } w = \text{side length of } v \cdot 2. \end{cases}$$

$$\{u, v\} \text{ is not well-separated} \rightarrow \text{Rep}(v) \cap \text{Ball}(w, (s+1)\text{rad}(w)) \neq \emptyset$$



By triangle inequality

To assert $v \cap \text{ball} \neq \emptyset$

Packing Lemma: $O\left(\frac{(2s+1)^d}{s^d}\right) = O(s^d) = O(1)$.

① u is a child of w (at most 4!)

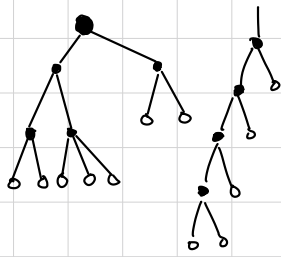
② v comes from $O(s^d)$ possible nodes

w can be charged $\leq 4 \cdot O(s^d) = O(s^d)$ times.

$$U \times V.$$

$\rightarrow$ The tree has at most $O(n \log \Delta)$ nodes $\rightarrow$ produces $O(n \log \Delta)$ pairs.

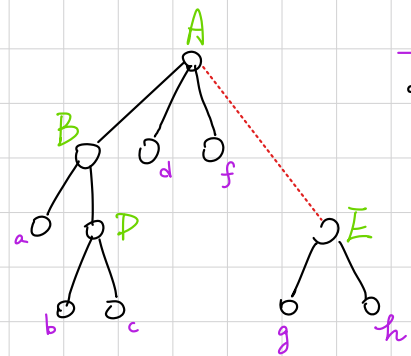
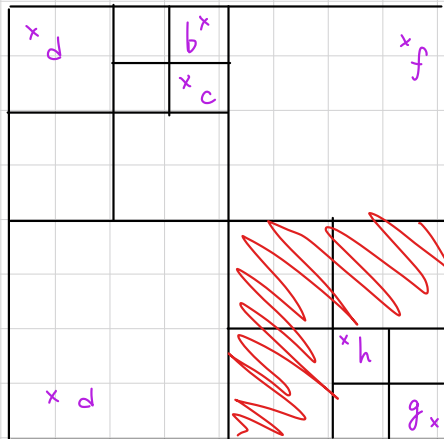
Path Compression of Quadtree



Every non-leaf-node has at least two children.

$\geq$ leaf nodes $\rightarrow$ # internal nodes $\leq n-1$.

Limiting # internal nodes.



1	3
2	4

$O(n \cdot \log \Delta)$ construction. Can take $O(n \log n)$ for granted.

Lecture 13- Applications of WSPD

• Diameter of P

$$\text{diam}(P) = d_{\max} = \max_{p, q \in P} \text{dist}(p, q)$$

• Convex Hull

$$2D \rightarrow O(n \log n)$$

$$3D \rightarrow O(n \log n)$$

THINK..

$$d \geq 4 \rightarrow O(n^2)$$

• Approximation Algorithm

1. Return $p, q \in P$

$$(1-\varepsilon) \text{diam } P \leq \text{dist}(p, q) \leq \text{diam } P.$$

$$O(n \log n + (\frac{1}{\varepsilon})^d \cdot n) \text{ for constant } \varepsilon.$$

Construct Quadtrees. All pairs

Proof. Let the WSPD of P be $\{\{A_1, B_1\}, \{A_2, B_2\}, \dots, \{A_m, B_m\}\}$, $m = O(n)$

Let $S = \{p_i, q_i : i \in [m], p_i, q_i \text{ is randomly picked from } A_i, B_i, \text{ resp.}\}$ representative.

Evaluate dist of these n pairs of points.

Let $p^*, q^* \in P$ st. $\text{dist}(p^*, q^*) = \text{diam}(P)$



$\exists i \in [1, m]$ st. $p^* \in A_i, q^* \in B_i$ (By definition)

$$\text{dist}(a_i, b_i) \geq \frac{s}{s+4} \text{dist}(p^*, q^*).$$

$$\text{Let } \frac{s}{s+4} = 1 - \varepsilon.$$

$$\Rightarrow s = \frac{4(1-\varepsilon)}{\varepsilon} = O(\frac{1}{\varepsilon}).$$

2. Spanner Graph

Let P : a set of n points in $\mathbb{R}^d$

$G = (V, E)$ st.

- $V = P$

- Each edge $\{u, v\} \in E$ carries weight $w(u, v) \triangleq \text{dist}(u, v)$.

T-spanner

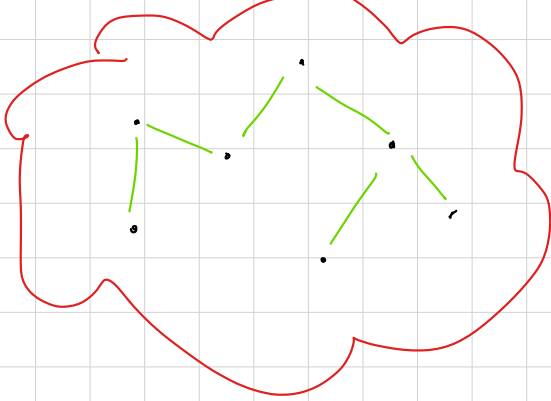
$G = (P, E')$ s.t. $\delta_G(p, q) \triangleq$ SP distance between p, q

G is a t -spanner if for all $p, q \in P$
 $\text{dist}(p, q) \leq \delta_G(p, q) \leq t \cdot \text{dist}(p, q)$.

→ Explain why $s = \frac{4(t+1)}{t-1}$

→ Explain the size of T -spanner

- Euclidean MST



$\text{Cost}(T)$ = the total length of all edges in T .

2D $\mathcal{O}(n \log n)$

3D $\mathcal{O}(n \log n)^{4/3}$

$\Omega(n^{4/3})$

$d \geq 4 \sim \mathcal{O}(n^2)$

T^* : an EMST

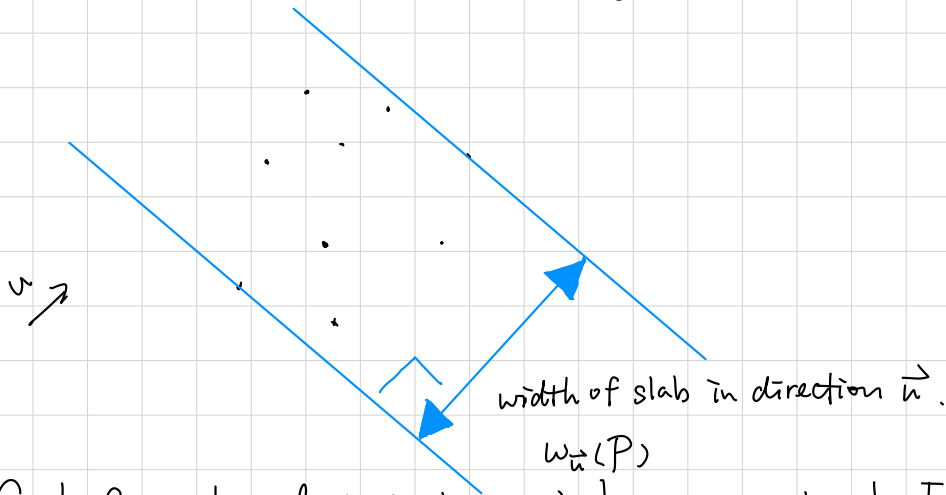
Goal : Find an EMST T s.t.

$\text{Cost}(T) \leq (1+\epsilon) \text{Cost}(T^*)$

$s = \frac{4(2+\epsilon)}{\epsilon} = \mathcal{O}(\frac{1}{\epsilon})$

Lecture 14 - (Grid Decomposition) ϵ -kernels

- Target: Coreset for solving a geometric problem approximately
 $\rightarrow$ Distributed Computing



- Goal: Given a directional vector $\vec{u}$, derive a value w st. $(1-\epsilon) w_{\vec{u}}(P) \leq w \leq w_{\vec{u}}(P)$
- ϵ -kernel: Find a subset $R \subseteq P$ st. $(1-\epsilon) w_{\vec{u}}(P) \leq w_{\vec{u}}(R) \leq w_{\vec{u}}(P)$

Claim. Exists a coreset with size $|R| = O((1/\epsilon)^d)$.

- Affine transformation.

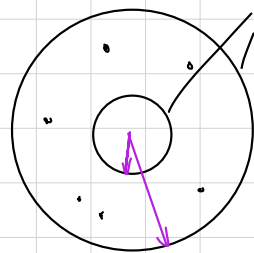
$\exists A \in \mathbb{R}^{d \times d}$ for mapping $\mathbb{R}^{d \times 1} \rightarrow \mathbb{R}^{d \times 1}$ st. $\vec{y}_{d \times 1} = A_{d \times d} \vec{x}_{d \times 1} + \vec{b}_{d \times 1}$.

Lemma: If R is an ϵ -kernel for P , any affine transformation of R $T(R)$ is an ϵ -kernel for $T(P)$.

- α -fat ($\alpha \leq 1$).

- $\mathcal{K}$: a convex body in $\mathbb{R}^d$

- $\mathcal{K}$ is α -fat if two concentric balls Π_1, Π_2

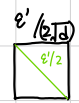


$$\alpha = \frac{\lambda_-}{\lambda_+}$$

- α -canonical position
 - K = a convex body in $\mathbb{R}^d$
 - K is in an α -canonical position if it is α -fat with $\lambda_+ = \frac{1}{2}$ and $\lambda_- = \frac{1}{2} \cdot \alpha$

- Lemma: There exists an affine transformation T such that $T(P)$ is $(1/d)$ -canonical.
 T can be found in $O(n)$ time.

Algorithm. (All points $p \in P$ are transformed s.t. $T(P)$ is $\frac{1}{d}$ -canonical, $w_{\vec{u}}(P) \leq 1, \forall |\vec{u}|=1$.)

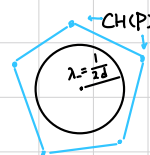
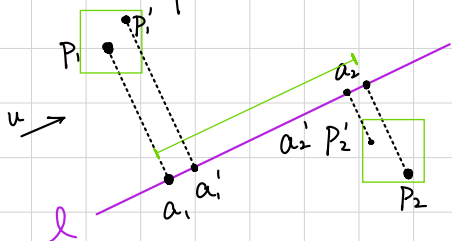
- ① $\varepsilon' = \varepsilon/d$ due to the lemma. diagonal $\frac{\varepsilon'}{2}$
- ② Impose a grid where each cell has side length $\varepsilon'/(2\sqrt{d})$ 
- ③ From every non-empty cell, take an arbitrary representative and form R .
 $O(n)$ time expected. (by hashing)
- ④ Size of R : $O((\frac{1}{\varepsilon'})^d)$. (by packing lemma, the ball with $r = \frac{1}{2}$ can cover at most $O(\frac{1}{\varepsilon'^d})$ cells)

Correctness.

Required to prove $(1-\varepsilon)W_{\vec{u}}(P) \leq W_{\vec{u}}(R) \leq W_{\vec{u}}(P)$

Fact: After the affine transformation, $w_{\vec{u}}(T(P)) \geq \frac{1}{d}$.

$$W_{\vec{u}}(P) = \max_{\vec{p} \in P} \vec{u} \cdot \vec{p} - \min_{\vec{p} \in P} \vec{u} \cdot \vec{p}. \quad \varepsilon \cdot W_{\vec{u}}(P) \geq \varepsilon/d.$$



Goal. Prove that $\text{dist}(a_1', a_2') \geq (1-\varepsilon)\text{dist}(a_1, a_2) = (1-\varepsilon)W_{\vec{u}}(P)$

Proof. $W_{\vec{u}}(R) \geq \text{dist}(a_1', a_2') \geq \text{dist}(a_1, a_2) - \text{dist}(p_1, p_1') - \text{dist}(p_2, p_2')$

By vector dot product $\vec{p}_1 \vec{p}_1' \cdot \vec{u} \leq |\vec{p}_1 \vec{p}_1'|, \vec{p}_2 \vec{p}_2' \cdot \vec{u} \leq |\vec{p}_2 \vec{p}_2'|$.

Since $|\vec{p}_1 \vec{p}_1'| \leq \varepsilon'/2, |\vec{p}_2 \vec{p}_2'| \leq \varepsilon'/2$.

$$W_{\vec{u}}(R) \geq \text{dist}(a_1, a_2) - \varepsilon' = W_{\vec{u}}(T(P)) - \varepsilon/d \geq (1-\varepsilon)W_{\vec{u}}(T(P))$$

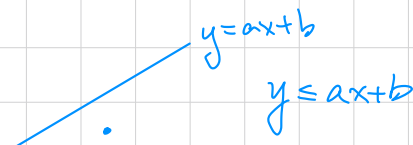
Lecture 15 - Point-Plane Duality

$$(a, b) \Leftrightarrow y = ax - b.$$

$$y = ax + b \Leftrightarrow (a, -b)$$

Primal

Dual



Denote p^* the dual of a point p , l^* the dual of a line l .

Basic properties.

① Incidence Preserving

$$p \text{ on } l \Leftrightarrow p^* \text{ on } l^*$$

Proof. $p = (a, b)$, $l: y = a_1x + b_1$

$$a_1a + b_1 = b. \quad \text{--- ①}$$

$$p^*: y = ax - b, l^* = (a_1, -b_1)$$

$\Rightarrow l^*$ is on p^* by ①.

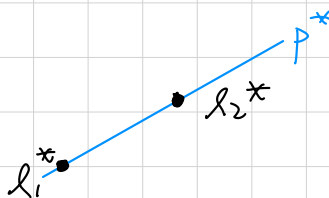
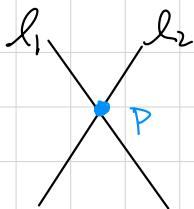
② Order reversal

$$p \text{ is above (below) } l \Leftrightarrow l^* \text{ is above (below) } p^*$$

(point above (below) line)

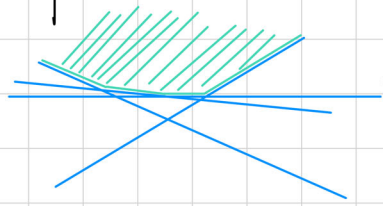
$$\text{Above: } y \geq ax + b$$

③ Intersection Preserving



Since $p \in l_1 \cap l_2$,
 $p \in l_1, p \in l_2$.
 Thus, $l_1^* \in p^*, l_2^* \in p^*$.

Envelopes

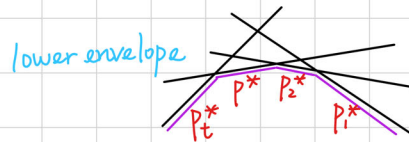
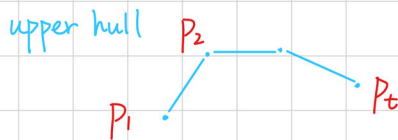


Given a set of n lines L in $\mathbb{R}^2$, the upper envelope is the intersection of the halfplanes

$$\begin{aligned} y &\geq a_1x + b_1 \\ y &\geq a_2x + b_2 \\ &\vdots \\ y &\geq a_nx + b_n \end{aligned}$$

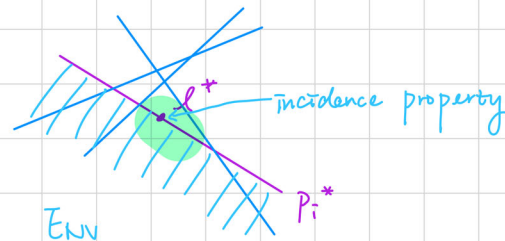
Lemma. (Convex Hull v.s. Envelopes)

points on upper hull $(L \rightarrow R)$ $\xLeftrightarrow{\text{duality}}$ lines on lower envelope $(R \rightarrow L)$



Proof. (Sketch)

① Appearance



② Ordering

$P_1 \rightarrow P_2 \rightarrow \dots \rightarrow P_t$ increasing x-coord

$P_1^* \rightarrow P_2^* \rightarrow \dots \rightarrow P_t^*$ increasing slopes



convex intersections

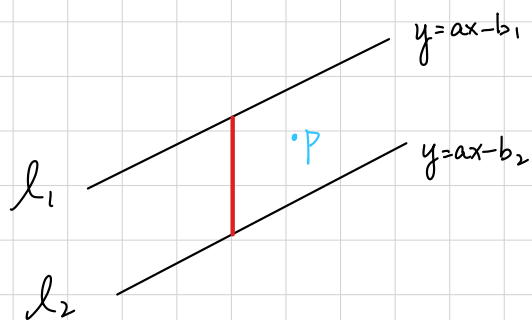
① any l must top p .

② any p^* must top l^*

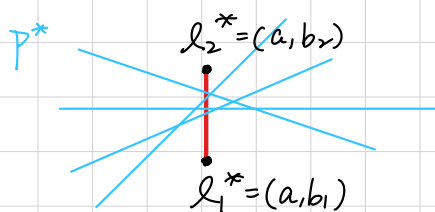
$\Rightarrow l^*$ must appear on the lower envelope.
vice versa.

Application:

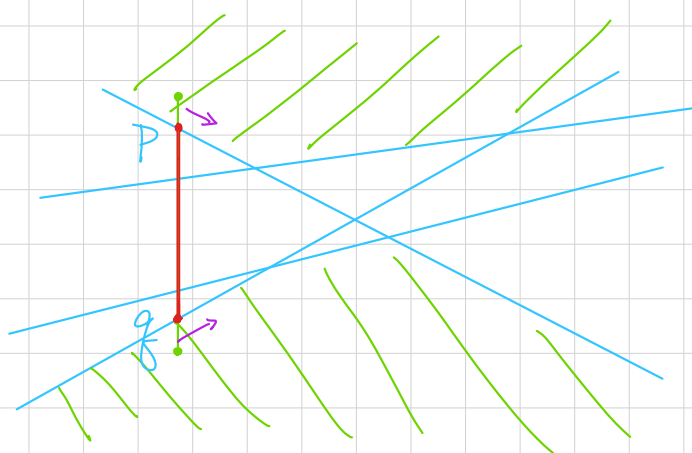
Find a slab with the smallest **vertical** distance containing P .



Primal space



Dual space



Must choose $p \in \partial(E)$, $q \in \partial(I)$

Must have at least one point on intersection on two lines.

$O(n \log n)$ planesweep.

perpendicular case: left as exercise

Extension: d -dimensional duality

Primal space: point $(p_1, p_2, \dots, p_d)$
(Dual) $\leftrightarrow$

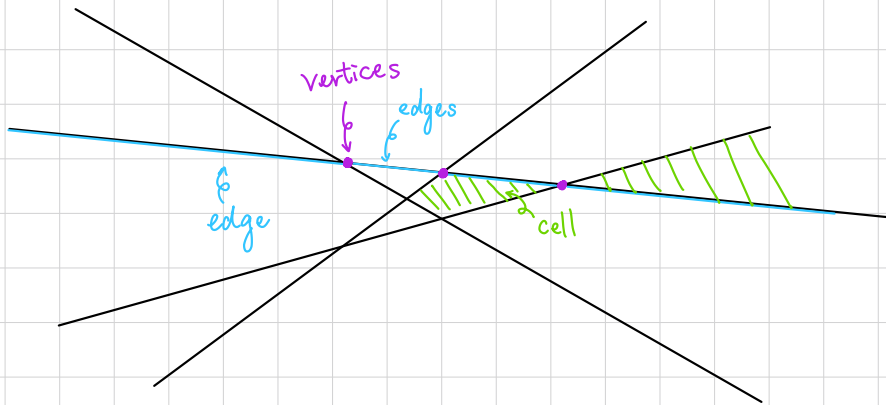
Dual space: hyperplane $y = p_1 x_1 + p_2 x_2 + \dots + p_{d-1} x_{d-1} - p_d$
(Primal)

Lecture 16 - Line Arrangements

Assumption: L is a set of n lines

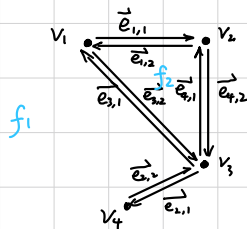
(non-parallel to each other, no three lines meet at a point)

$A(L)$: an arrangement of L .



For n lines,
each line is chopped into
 n subsegments.

- Vertex record
 - $\text{Coordinate}(v)$
 - $\text{IncidentEdge}(v)$: pointing to an arbitrary edge having v as origin
- Face record
 - $\text{OuterComponent}(f)$: pointing to some half-edge on its outer boundary
 - $\text{InnerComponent}(f)$: pointing to some half-edge on the boundary of holes
- Half-edge record
 - $\text{Origin}(e)$
 - $\text{Twins}(e)$
 - $\text{IncidentFace}(e)$: pointing to the bounding face
always on observer's left



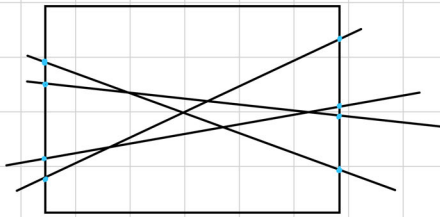
Vertex	Coordinates	Incident Edge
v_1	(0, 4)	$\vec{e}_{1,1}$
v_2	(2, 4)	$\vec{e}_{4,2}$
v_3	(2, 2)	$\vec{e}_{2,1}$
v_4	(1, 1)	$\vec{e}_{2,2}$

Face	OuterComponent	InnerComponents
f_1	nil	$\vec{e}_{1,1}$
f_2	$\vec{e}_{4,1}$	nil

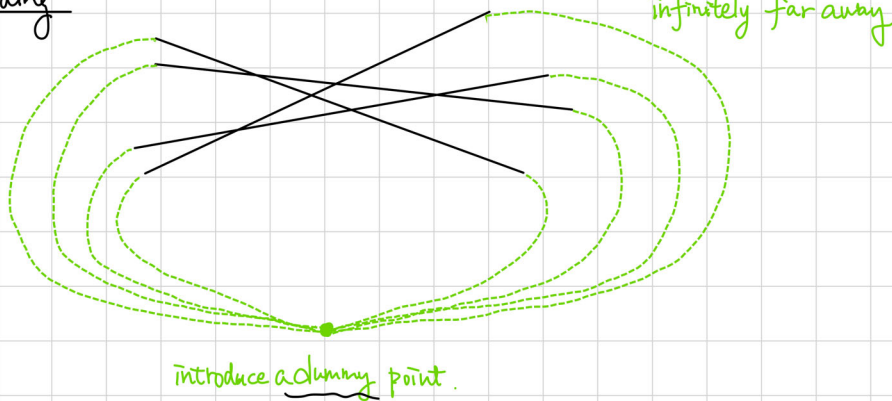
Half-edge	Origin	Twins	IncidentFace	Next	Prev
$\vec{e}_{1,1}$	v_1	$\vec{e}_{1,2}$	f_1	$\vec{e}_{4,2}$	$\vec{e}_{3,1}$
$\vec{e}_{1,2}$	v_2	$\vec{e}_{1,1}$	f_2	$\vec{e}_{3,2}$	$\vec{e}_{4,1}$
$\vec{e}_{2,1}$	v_3	$\vec{e}_{2,2}$	f_1	$\vec{e}_{2,2}$	$\vec{e}_{4,2}$
$\vec{e}_{2,2}$	v_4	$\vec{e}_{2,1}$	f_1	$\vec{e}_{3,1}$	$\vec{e}_{2,1}$
$\vec{e}_{3,1}$	v_3	$\vec{e}_{3,2}$	f_1	$\vec{e}_{1,1}$	$\vec{e}_{2,2}$
$\vec{e}_{3,2}$	v_1	$\vec{e}_{3,1}$	f_2	$\vec{e}_{4,1}$	$\vec{e}_{1,2}$
$\vec{e}_{4,1}$	v_3	$\vec{e}_{4,2}$	f_2	$\vec{e}_{1,2}$	$\vec{e}_{3,2}$
$\vec{e}_{4,2}$	v_2	$\vec{e}_{4,1}$	f_1	$\vec{e}_{2,1}$	$\vec{e}_{1,1}$

Rigorous fixes

Box imposing.



Banding



$\mathcal{O}(n^2)$ vertices, edges, cells

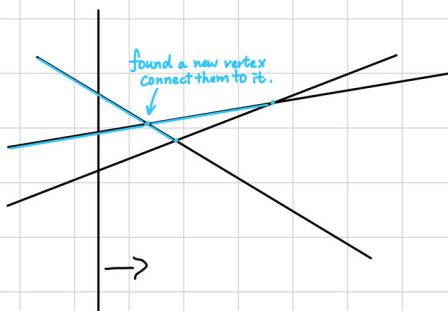
Proof.

$$V = \binom{n}{2}, E = n^2, F = 2 + E - (V + 1) = 1 + \binom{n}{2} + n.$$

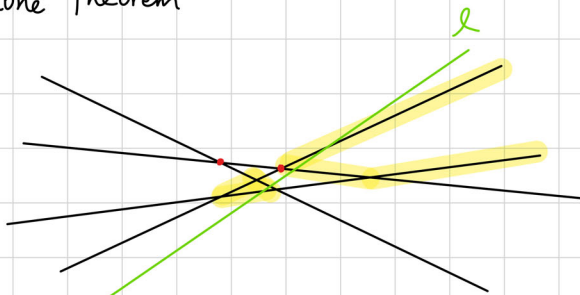
- Computing vertices $\sim \mathcal{O}(n^2)$

Enumerate any two lines l_1, l_2 .

- Computing cells $\sim \mathcal{O}(n^2 \log n)$



- Zone Theorem



For any line l , zone complexity $\leq 6n$.

Zone: the set of cells intersecting with l

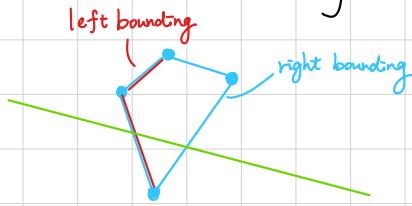
Zone complexity: total # of edges of the cells in the zone.

$(i+1)^{\text{st}}$ line: zone complexity $\mathcal{O}(i)$

$$\Rightarrow \sum_{i=1}^n \mathcal{O}(i) = \mathcal{O}(n^2), \quad (\text{Brute force is optimal})$$

Lecture 17 - Proof of Zone Theorem

suffices to prove that #left bounding edges $\leq 3 \cdot i$

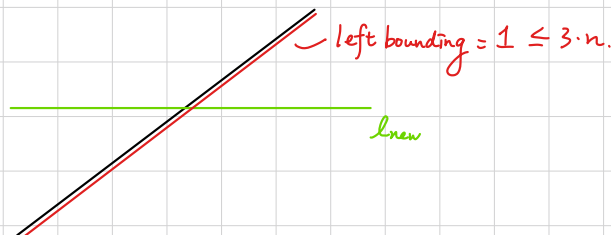


Generalization
of position

Rotate the grid so that the new line inserted is horizontal

Proof. By induction.

(i) (Base Case) when $n=1$,



(ii) (Strong Inductive Step) suppose the statement hold for $n \leq k$.
prove for $n = k+1$.



Let l^* be the line producing the rightmost intersection point with l .

Consider the removal of l^* . there are k lines left.

Apply the induction hypothesis, # left bounding edges in the zone $\leq 3k + \text{last cell}$.

Considering traversing along l_{ins} from left to right on the graph after removal

a. unbounded from the right

b. number of left bounding edges at most 3

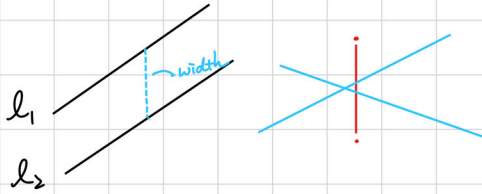


Lecture 18 - Line Arrangements and Applications

- k -corridor

P : a set of n points in $\mathbb{R}^2$.

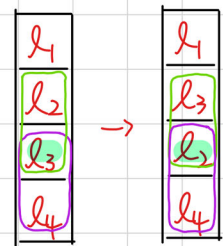
Find a corridor with the smallest width that encloses k points of P .



By planesweep in $O(n^2 \log n)$, $O(n)$ space.

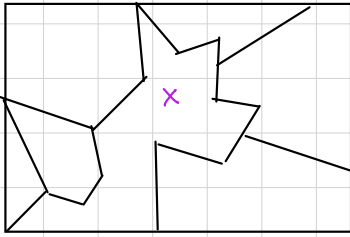
Maintain an array of the ordering of these k lines.

Two lines gets swapped at every intersection point.



Example: 2-corridor problem

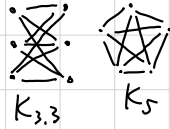
Lecture 19 - (Planar Graphs) Point Location



subdivision

face

planar graph: a graph that can be "drawn" s.t. no two edges cross each other

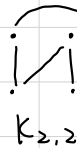


$K_{3,3}$

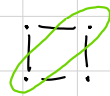
K_5

a graph is planar iff it does not have a subgraph isomorphic to $K_{3,3}$ or K_5 .

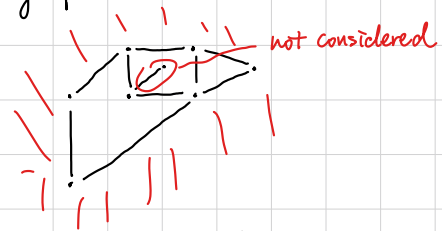
SLPG: short line planar graph



$K_{2,2}$



(SLPG)



Regular-SLPG: every edge in the planar graph bounds two distinct faces.

Euler's formula: For every connected SLPG.

$$\# \text{ vertex} - \# \text{ edge} + \# \text{ face} = 2$$

Triangulation region complexity $O(n)$. Changing arg.

1. Each point changed twice
2. Each polygon has $n-3$ diagonals

Bounding the planar graph with triangle $\rightarrow$ rectangle
 $\rightarrow$ bound the rectangle

Lemma (Independent Set of SLPG): For any triangulated SLPG with $n \geq 12$ vertices. There must exist an independent set S st.

can be removed

- S does not contain the 3 dummy points
- $|S| \geq \frac{n}{48}$
- Every vertex in S has a degree at most 11.

Proof. In an triangulated SLPG, $m = 3n - 6$.

$$\sum_{v \in V} \deg(v) = 2m = 6n - 12 < 6n.$$

Claim: There are $\geq \frac{n}{2}$ vertices having degree ≤ 11 .

Suppose the otherwise that $\geq \frac{n}{2}$ vertices having degree ≥ 12

Then, $\sum_{v \in V} \deg(v) \geq 6n$, a contradiction.

- ① Eliminate all vertices with degree ≥ 12 and also the outer three dummy points (at most $\frac{n}{2} - 3 \geq \frac{n}{4}$ of them)
- ② Take an uneliminated vertex into S
 - Eliminate its neighbors (at most 11)
 - Eliminate itself

$\rightarrow$ Each step eliminates at most 12 points corresponding to 1 vertex in S

Graph Sparsification

G_0 = original SLPG

S = independent set of G_0

Removal of S in G_0

Let G_1 = triangulated $G_0/S \leq \frac{47}{48}$ vertices

G_0

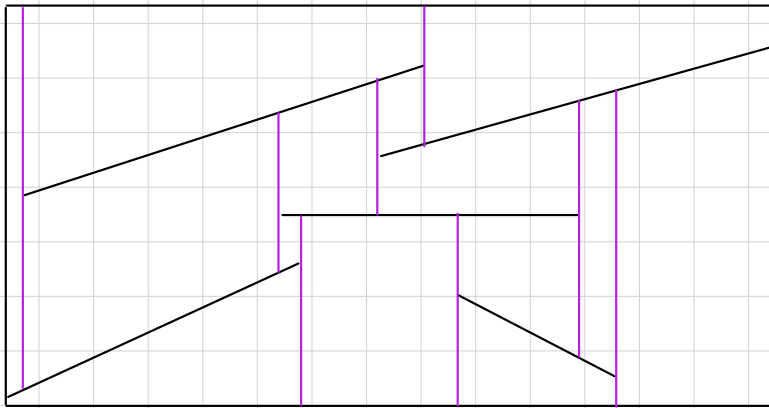
$\Downarrow$

G_1

$\Downarrow$

G_h ($h = \log n$)

Kirkpatrick's point location algorithm



Trapezoidal Map

Complexity:

1. at most $O(n)$ rays
2. a ray can break an existing line into two more pieces

Lecture 20 - Properties of Voronoi Diagrams

P : a set of n points in $\mathbb{R}^2$.

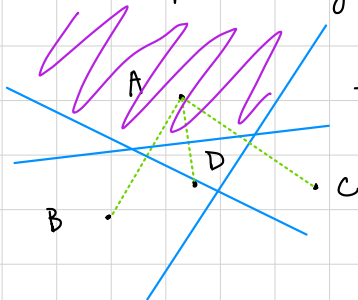
General position assumption

1. No 3 points in P are on the same line
2. No 4 points in P are on the same circle

- Nearest Neighbour

Given a query point $q \in \mathbb{R}^2$. return a point $p \in P$ with the smallest distance $\text{dist}(p, q)$.

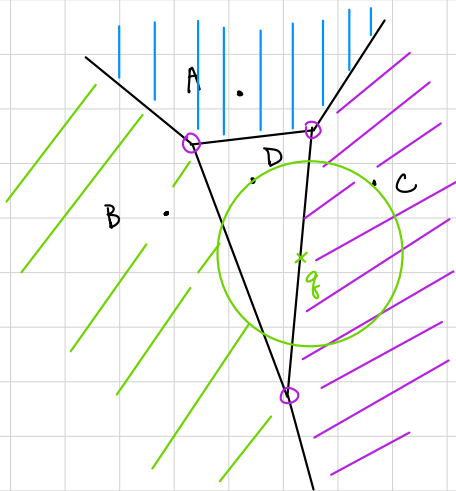
Goal: $O(n)$ space, $O(\log n)$ time.



→ What is the set of points which can have point A as the nearest neighbour?

A polygon containing A (called Voronoi cell of A , $V(A) = \{q \in \mathbb{R}^2 \mid A \text{ is a NN of } q\}$)

Voronoi Diagram: $\bigcup_{p \in P} V(p)$.



Fact 1. A point q is on an edge $\Leftrightarrow \exists$ a circle centered at q passing ≥ 2 points in P and the circle is empty in the interior

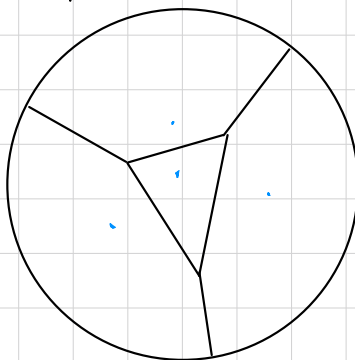
Fact 2. A point q is on the vertex $\Leftrightarrow \exists$ a circle centered at q passing ≥ 3 points in P and the circle is empty in the interior

Fact 3. Every vertex in the Voronoi diagram has degree of 3.
(Under general position)

Theorem. Voronoi diagram of a point set P has complexity $O(n)$.

→ Implication: Nearest neighbor search is solved in $O(\log n)$ time (point location)

Proof.



$$\sum_{v \in V} \deg(v) = 3V = 2E.$$

$$V - E + F = 2.$$

$$0.5V = f - 2$$

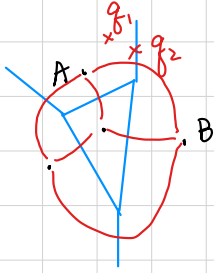
$$V = 2f - 4 = O(f).$$

interior faces: corresponds to every Voronoi cell. $O(n)$.

exterior face: bounded by circle.

$$f = n + 1.$$

Lecture 21 - Properties of Delaunay Triangulations



We can always adjust the red edges within the two regions s.t. no two edges intersects.

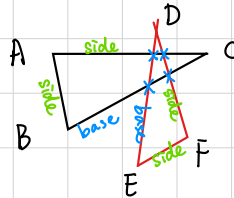
Want to straighten the lines
→ replacement? (Warn: planar property)

Lemma: Every point set triangulation has $2n-2-2$ edges.

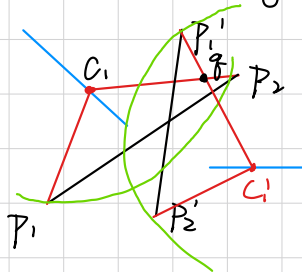
- Circumcircle property: Circumcircle of any triangle in the Delaunay triangulation is empty.
- Empty circle property: two sites p_i, p_j are connected by an edge in the Delaunay triangulation, iff there is an empty circle passing through p_i and p_j .
- Closest pair property: the closest pair of sites in P are neighbors in the Delaunay triangulation.
- Bijection between Voronoi Cell to Delaunay triangle (Duality)
- $O(n \log n)$ precise output for EMST.

Claim: Let $\overline{p_1 p_2}$, $\overline{p'_1 p'_2}$ be two red lines. Prove that: The line segments $\overline{p_1 p_2}$ and $\overline{p'_1 p'_2}$ do not intersect with each other.

- ① BC intersects with DE
- ② $\triangle ABC$ does not cover D.E.
 $\triangle DEF$ does not cover B.C.



Then, at least one side edge of ABC intersects with one side edge of DEF.

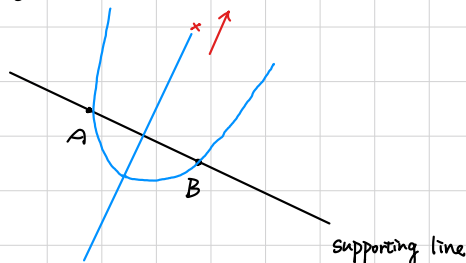


$\triangle C_1 p_1 p_2$ does not contain p'_1, p'_2

$\triangle C'_1 p'_1 p'_2$ does not contain p_1, p_2

Contradiction: Voronoi cells are disjoint while q lies in both cells.

- Every edge of $\text{conv}(P)$ is an edge in $\text{DT}(P)$



Lecture 22 - Backward Analysis

- Randomized algorithms

A = an event

$\mathbb{P}(A)$: probability that event A happens

$B_1, B_2, \dots, B_t$: disjoint events covering the whole probability space

Total probability law: $\mathbb{P}(A) = \sum_{i=1}^t \mathbb{P}(A|B_i) \mathbb{P}(B_i)$.

- Example: Bounding Interval Problem

Algorithm:

$A' \leftarrow$ a random permutation of A

$x \leftarrow \min\{A'[1], A'[2]\}$

$y \leftarrow \max\{A'[1], A'[2]\}$

for $i = 3$ to $|A|$ do

if $A'[i] \notin [x, y]$

$\mathcal{O}(i)$ $\left\{ \begin{array}{l} x = \min\{A'[1], A'[2], \dots, A'[i]\} \\ y = \max\{A'[1], A'[2], \dots, A'[i]\} \end{array} \right.$

Scan

return $[x, y]$.

Algorithm permute /* generate a random permutation of A */

for $i \leftarrow 2$ to n do

$x \leftarrow \text{randint}(1, i)$

swap($A[i], A[x]$)

Fisher-Yate's shuffle

Proof:

$$\mathbb{P}(\Pi = \pi) = \mathbb{P}(x'[1] = b_1) \cdot \mathbb{P}(x'[2] = b_2 | x'[1] = b_1) \cdot$$

$$\mathbb{P}(x'[3] = b_3 | x'[1] = b_1, x'[2] = b_2) \cdot \dots$$

$$\mathbb{P}(x'[N] = b_N | x'[1] = b_1, x'[2] = b_2, \dots, x'[N-1] = b_{N-1})$$

$$= \prod_{i=1}^N \frac{1}{i} = \frac{1}{N!}.$$

$$\mathbb{E}[\text{running time}] = \mathcal{O}(n) + \sum_{i=3}^n [\mathcal{O}(1) + \mathbb{P}(X_i = 1) \cdot \mathcal{O}(i)]$$

Claim: $\mathbb{P}(X_i = 1) \leq 2/i$. X_i is a random variable, $X_i = \begin{cases} 1 & \text{if the interval changed} \\ 0 & \text{if the interval does not change} \end{cases}$

Obs. $X_n = 1$ iff $A'[n]$ is min/max in A'

$$\mathbb{P}(X_n = 1) = 2/n.$$

Consider any $i < n$

Fact. $X_i = 1$ iff $A'[i]$ is max/min in $A'[1], A'[2], \dots, A'[i]$.

Define $\Pi = (A[i+1], A[i+2], \dots, A[n])$.

$$\mathbb{P}(\Pi = \pi) = \frac{1}{\binom{n}{n-i} (n-i)!} = \frac{2!}{n!}$$

$$\mathbb{P}(X_i = 1) = \sum_{\text{all } \pi} (\mathbb{P}(X_i = 1 | \Pi = \pi) \cdot \mathbb{P}(\Pi = \pi)) \quad (\text{backward analysis})$$

$$= \sum_{\text{all } \pi} 2/i \cdot \frac{i!}{n!}$$

$$= \frac{n!}{i!} \cdot \frac{i!}{n!} \cdot 2/i$$

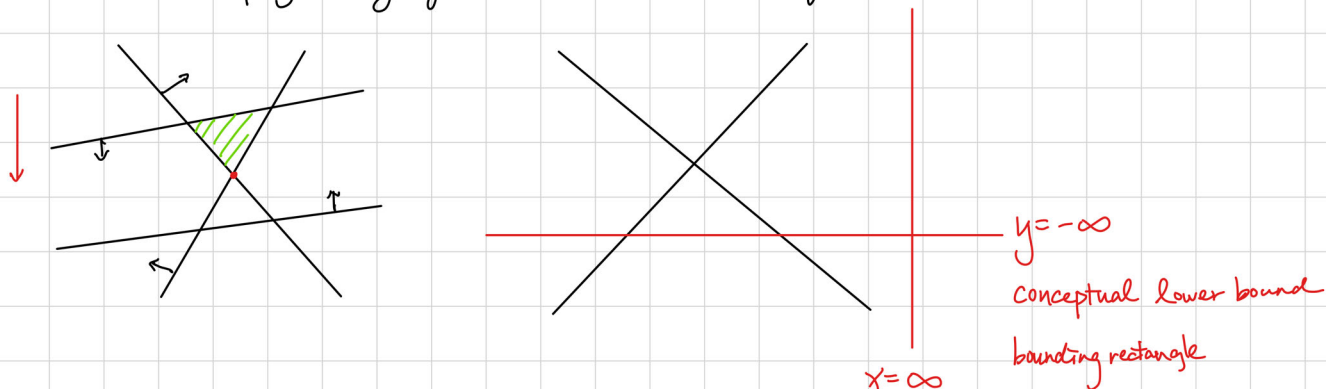
$$= 2/i.$$

Lecture 23 - (Backward Analysis) Linear Programming

H : a set of n halfplanes in $\mathbb{R}^2$

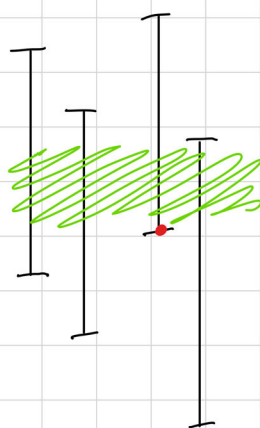
$\cap H$: the polygonal intersection of n halfplanes

Linear programming goal: Find the smallest y -coordinate of $\cap H$



General position assumption: There is no three lines passing through the same point.

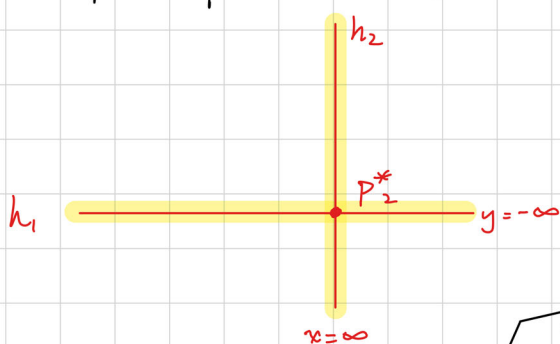
1D linear programming:



$\mathcal{O}(1)$ per interval

$\mathcal{O}(n)$ time for n intervals

Let $H_1, H_2, \dots, H_n$ be n halfplanes, $\cap H_i$: the intersection of $H_1, H_2, \dots, H_i$.
 p_i^* : optimal point on $\cap H_i$.



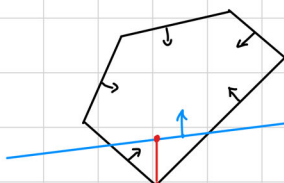
Consider $i \geq 3$, p_{i-1}^* : available Goal = p_i^* .

Claim: ① p_{i-1}^* satisfies H_i , $p_i^* = p_{i-1}^*$

Proof. Suppose $p_i^* \neq p_{i-1}^*$. $p_i^* < p_{i-1}^*$, but then p_{i-1}^* can be as small as p_i^* .

② otherwise, p_{i-1}^* must fall on the other side of the current considered halfplane.

Suppose p_i^* is not on the supporting line of the halfplane, connect p_i^* , p_{i-1}^* and investigate the point of intersection p' . By convexity, p' must be better than p_i^* a contradiction.



Incremental construction

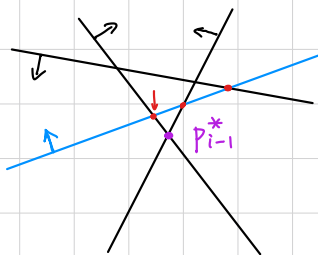
① Permutate n halfplanes

② Initial scenario: bounded two halfplanes $x \leq \infty, y \geq -\infty$

p_{i-1}^*

if $p_{i-1}^* \in H_i$ $p_i^* = p_{i-1}^* \dots \mathcal{O}(1)$

else

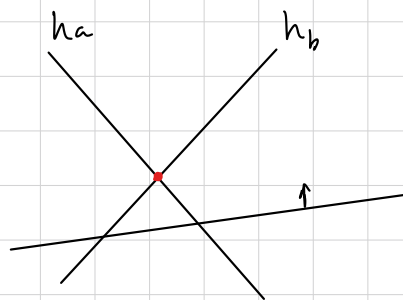


1D instance: on the new line

Work: apply backward analysis.

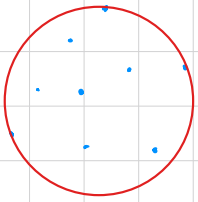
$\mathbb{P}(p_i^* \text{ is different from } p_{i-1}^*) \leq d/n.$

$\mathcal{O}(d!n)$. only applies for low-dimensionality



Lecture 24 - (Backward Analysis) Minimum Enclosing Ball

P : a set of n points in $\mathbb{R}^d$.

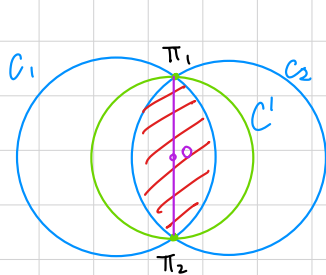


Solvable in $O(n)$ expected time.

Focus on $d=2$. General position assumption: No 4 co-circular points

Lemma 1. The minimum enclosing ball B is unique.

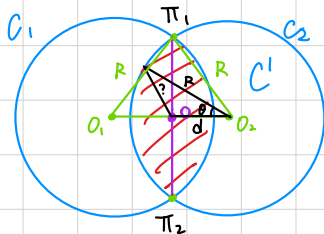
Proof. Suppose, on the contrary that there are two circles C_1, C_2 with radius R .



Since $P \subseteq C_1, P \subseteq C_2, P \subseteq C_1 \cap C_2$.

Let π_1, π_2 be the intersection points of the two circles.

Consider the circle C' centered at the midpoint o of $\pi_1 \pi_2$ and having radius half of $\pi_1 \pi_2$.
covers the area of intersection (hence also P) but is smaller than C_1, C_2 which is a contradiction.



Reasoning: Let the distance between the centers of the two circles O_1, O_2 be $2d$.

$$\because O_1 \pi_1 = O_2 \pi_1 = R \quad \therefore \overline{O_1 o} = \overline{O_2 o} \quad (\text{by } \triangle O_1 \pi_1 o \cong \triangle O_2 \pi_1 o)$$

$$\overline{O_1 o} = \overline{O_2 o} = d$$

$$\Rightarrow \overline{\pi_1 o} = \overline{\pi_2 o} = \sqrt{R^2 - d^2} \geq \sqrt{d^2 + R^2 - 2Rd \cos \theta}, \cos \theta \in \left[\frac{d}{R}, 1\right].$$

$= |op|$ where p is any point on the boundary of $C_1 \cap C_2$.

Proof (3-dimensional). Suppose that there are two balls B_1, B_2 with radius R .

Since $P \subseteq B_1, P \subseteq B_2, P \subseteq B_1 \cap B_2$.

Connect the center of the two balls, i.e. segment $\overline{C_1 C_2}$ with length $2d$ and consider the midpoint o .

Create a new ball B' centered at o with radius $\sqrt{R^2 - d^2}$.

B' covers the area of intersection (hence P) but is smaller than B_1, B_2 , which is a contradiction.

Reasoning: Consider the slice of B_1, B_2 intersecting with $z=k$ and note that the center's relative position does not change on the projection and thus apply the reasoning for $d=2$.

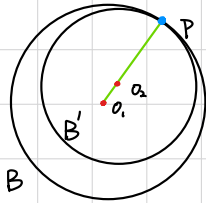
It is sufficient to consider the projection where the two circles have radius R .

Lemma 1. The boundary of $MEB(P)$ passes at least 2 points of P .

Proof. Case 1. passing 0 point

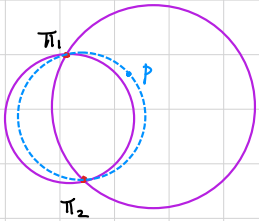
Given an arbitrary circle, we can always apply an infinitesimally small shrink on the circle, giving a contradiction.

Case 2. passing 1 point



Move the center along $\overline{o_1P}$ by an infinitesimally small amount with radius $\overline{o_2P}$.
 $\overline{o_2P} < \overline{o_1P}$ and B' is smaller than B , giving a contradiction

Lemma 2. Let C_1, C_2 be two intersecting circles such that $r(C_1) < r(C_2)$. Let L be the area inside both circles. Consider an arbitrary point p covered by C_2 but not C_1 . Then, there exist a circle C' smaller than C_2 , passes through P and covers L .



Proof. Consider the following fact: Fix two distinct points π_1, π_2 . Consider all circles passing through π_1 and π_2 . The centers of those circles must be on the perpendicular bisector of the segment $\overline{\pi_1\pi_2}$. Every such circle C can be divided into a ⁽ⁱ⁾left arc, which is the part of C on left of $\overline{\pi_1\pi_2}$, and ⁽ⁱⁱ⁾a right arc, which is the part of C on the right of $\overline{\pi_1\pi_2}$. As the center o of C moves away from the midpoint m of the segment $\overline{\pi_1\pi_2}$ towards right, the left arc morphs toward $\overline{\pi_1\pi_2}$, while the right arc morphs away from $\overline{\pi_1\pi_2}$; furthermore, C grows continuously.

Let π_1, π_2 be the intersection points of C_1 and C_2 . Imagine "morphing" a circle C from C_2 to C_1 while ensuring that C passes π_1, π_2 . Stop as soon as the right arc of C hits p . As C_1 is no larger than C_2 , C must be smaller than C_2 . Thus, C is the circle that we are looking for.

Proof. (3-dimensional) Every ball can be divided into two curved surfaces on left/right of the intersection (i.e. a circle) of two balls, which also behaves as the 2D case.

Then, we morph a ball B from B_2 to B_1 ensuring that B passes C , and stop as soon as B hits p .

- Two points are known

Algorithm 1: Two-points-fixed-MEB($P, \{p_1, p_2\}$)

$B \leftarrow$ the smallest ball enclosing p_1, p_2

$(p_3, p_4, \dots, p_n) \leftarrow$ an arbitrary permutation of the other points in P

for $i \leftarrow 3$ to n do

if $p_i \notin B$ then

$B \leftarrow$ the ball whose boundary passes p_1, p_2 and p_i

return B

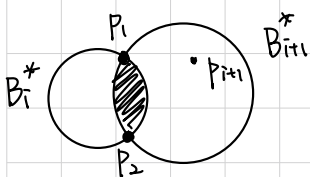
Lemma 3. Define $P_i = \{p_1, p_2, \dots, p_i\}$ for each $i \in [n]$. For any $i \in [2, n]$, define $B_i^* = \text{MEB}(P_i, \{p_1, p_2\})$.

If $p_{i+1} \in B_i^*$, then $B_{i+1}^* = B_i^*$. Otherwise, the boundary of B_{i+1}^* must pass p_{i+1} .

Proof of correctness. If $p_{i+1} \in B_i^*$, then $B_{i+1}^* = B_i^*$ follows from the uniqueness of $\text{MEB}(P_{i+1}, \{p_1, p_2\})$.

Next, we consider $p_{i+1} \notin B_i^*$. Assume on the contrary that the boundary of B_{i+1}^* does not pass p_{i+1} .

Then, p_{i+1} falls in the interior B_{i+1}^* . The radius of B_{i+1}^* cannot be smaller than that of B_i^* (both of them cover P_i and pass p_1 and p_2 , but B_i^* is $\text{MEB}(P_i, \{p_1, p_2\})$). The entire P_i must fall in the intersection of B_i^* and B_{i+1}^* (shaded area). By Lemma 2, there is a ball smaller than B_{i+1}^* covering P_{i+1} and passing p_1, p_2 , which is a contradiction.



- One point is known

Algorithm 2: One-point-fixed-MEB($P, \{p_1\}$)

$(p_2, p_3, \dots, p_n) \leftarrow$ an arbitrary permutation of the other points in P

$B \leftarrow$ the smallest ball enclosing p_1, p_2

for $i \leftarrow 3$ to n do

if $p_i \notin B$ then

$B \leftarrow \text{Two-points-fixed-MEB}(\{p_1, p_2, \dots, p_i\}, \{p_1, p_i\})$

return B

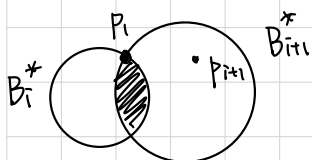
Lemma 3. Define $P_i = \{p_1, p_2, \dots, p_i\}$ for each $i \in [n]$. For any $i \in [2, n]$, define $B_i^* = \text{MEB}(P_i, \{p_1\})$.

If $p_{i+1} \in B_i^*$, then $B_{i+1}^* = B_i^*$. Otherwise, the boundary of B_{i+1}^* must pass p_{i+1} .

Proof of correctness. If $p_{i+1} \in B_i^*$, then $B_{i+1}^* = B_i^*$ follows from the uniqueness of $\text{MEB}(P_{i+1}, \{p_1\})$.

Next, we consider $p_{i+1} \notin B_i^*$. Assume on the contrary that the boundary of B_{i+1}^* does not pass p_{i+1} .

Then, p_{i+1} falls in the interior B_{i+1}^* . The radius of B_{i+1}^* cannot be smaller than that of B_i^* . The entire P_i must fall in the intersection of B_i^* and B_{i+1}^* (shaded area). By Lemma 2, there is a ball smaller than B_{i+1}^* covering P_{i+1} and passing p_1 , which is a contradiction.



Other than p_1 , the boundary of B_{i-1}^* must pass at least one more point in P_i . No more than two points (under general position assumption).

Case (i): B_{i-1}^* passes one more point $\pi_1 \in P_i$. $p_i \notin B_{i-1}^*$ occurs only when $p_i = \pi_1$ with probability $\frac{1}{i-1}$.

Case (ii): B_{i-1}^* passes two more points $\pi_1, \pi_2 \in P_i$. $p_i \notin B_{i-1}^*$ occurs only when $p_i = \pi_1$ or π_2 with probability $\frac{2}{i-1}$.

By backward analysis, $E[T_i] = O(1)$, where T_i is time spent on i -th point.

- No point is known

Algorithm 3: MEB(P)

$(p_1, p_2, \dots, p_n) \leftarrow$ an arbitrary permutation of P

$B \leftarrow$ the smallest ball enclosing p_1, p_2

for $i \leftarrow 3$ to n do

if $p_i \notin B$ then

$B \leftarrow \text{Two-points-fixed-MEB}(\{p_1, p_2, \dots, p_i\}, \{p_i\})$

return B

Lemma 3. Define $P_i = \{p_1, p_2, \dots, p_i\}$ for each $i \in [n]$. For any $i \in [2, n]$, define $B_i^* = \text{MEB}(P_i)$.

If $p_{i+1} \in B_i^*$, then $B_{i+1}^* = B_i^*$. Otherwise, the boundary of B_{i+1}^* must pass p_{i+1} .

Proof of correctness. If $p_{i+1} \in B_i^*$, then $B_{i+1}^* = B_i^*$ follows from the uniqueness of $\text{MEB}(P_{i+1})$.

Next, we consider $p_{i+1} \notin B_i^*$. Assume on the contrary that the boundary of B_{i+1}^* does not pass p_{i+1} .

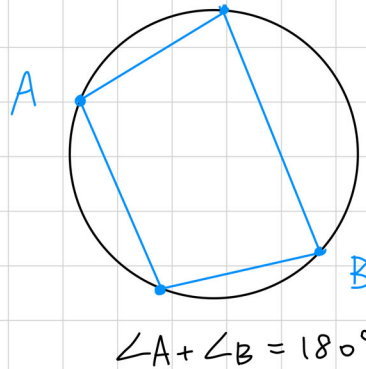
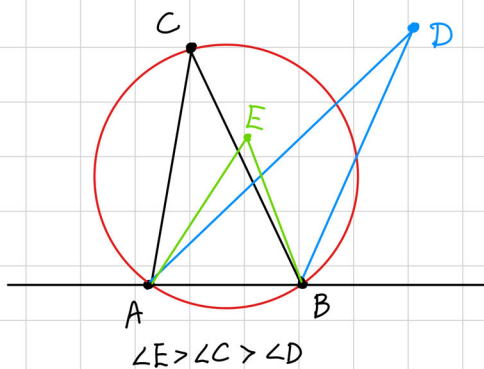
Then, p_{i+1} falls in the interior B_{i+1}^* . The radius of B_{i+1}^* cannot be smaller than that of B_i^* .

The entire P_i must fall in the intersection of B_i^* and B_{i+1}^* (shaded area). By Lemma 2, there is a ball smaller than B_{i+1}^* covering P_{i+1} , which is a contradiction.

By backward analysis. Algorithm 3 runs in expected $\mathcal{O}(n)$ time.

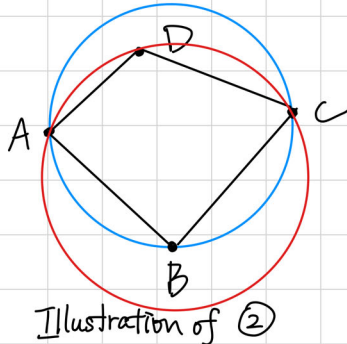
Lecture 25 - (Backward Analysis) Delannay Triangulation

Geometric Basis



Lemma: For any convex quadrilateral $ABCD$,
 ① D is in $\odot ABC$ iff B is in $\odot ABD$
 ② D is outside of $\odot ABC$ iff A is outside of $\odot BCD$

Proof.

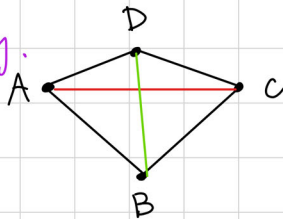


① D is inside of $\odot ABC$.
 $\iff \angle B + \angle D < 180^\circ$
 B is outside of $\odot ACD$.

② D is outside of $\odot ABC$.
 $\iff \angle B + \angle D > 180^\circ$
 B is outside of $\odot ACD$.

Edge Legality: An edge AC is illegal iff D lies inside $\odot ABC$ (or equivalently B in $\odot ACD$).

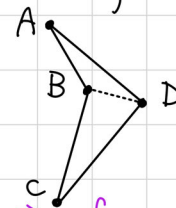
Corollary.



Each convex quadrilateral has exactly one legal diagonal and one illegal diagonal.

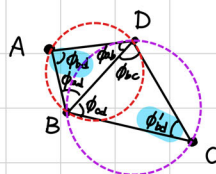
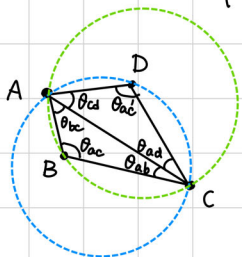
Example. DB : legal
 AC : illegal

concave quadrilateral
 only one possible diagonal



Lemma. An edge flip operation converts a triangulation of a convex quadrilateral from illegal to legal.

Let $\triangle ABC$ and $\triangle CDA$ be two adjacent triangles, such that their union forms a convex quadrilateral $ABCD$. Suppose that the initial triangle pair violates the empty circle condition, that point d lies inside the circumcircle of $\triangle ABC$ (which implies that B lies inside the circumcircle of $\triangle CDA$.)



$$\phi'_{bd} > \phi_{ad}, \phi_{ab}$$

$$\phi_{bd} > \phi_{cd}, \phi_{bc}$$

Since ϕ_{bd}, ϕ'_{bi} are larger than some angles from $\phi_{ab}, \phi_{bc}, \phi_{cd}, \phi_{da}$.

ϕ_{bd}, ϕ'_{bi} must be larger than the minimum of the four.

D lies inside of $\odot ABC$, $\phi_{bc} < \phi'_{bc}$, $\phi_{ab} < \phi'_{ab}$
 B lies inside of $\odot ACD$, $\phi_{ad} < \phi'_{ad}$, $\phi_{cd} < \phi'_{cd}$.

Therefore, flipping the edge from $\overline{AC}$ to $\overline{BD}$ will give a legal triangulation of the quadrilateral $ABCD$.

Triangulation of a set of points P in a plane is a graph having P as set of vertices which is rectilinear, planar and maximal in the number of edges.

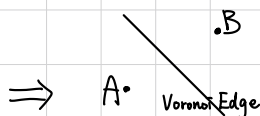
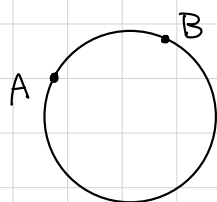
- Every triangulation of a point set has $2n-2-h$ triangles and $3n-h-3$ edges.
- Any triangulation of P contains the edges from the convex hull of P

Theorem. G is a legal triangulation of P in general position (no four points are on the same circle) if and only if G is the Delaunay triangulation of P .

($\Leftarrow$): No point is contained inside the circumcircle of any triangular face.
Thus, all edges are legal.

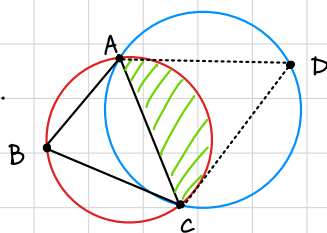
($\Rightarrow$):

Fact 1.



If an empty circle passing A and B exists, there is a Voronoi edge between A and B (and thus an edge connecting A and B in the Delaunay triangulation)

Fact 2.



Let D be a point outside $\odot ABC$ such that points B and D fall on different sides of the line passing through $\overline{AC}$. Then, $\odot ACD$ covers the shaded territory of arc $\widehat{AC}$ inside $\odot ABC$.

Apparently, $\widehat{AC}$ of $\odot ACD$ bounds $\overline{AC}$ from the left since $\overline{AC}$ divides $\odot ACD$ into left and right bounding arcs.

$\widehat{AC}$ of $\odot ABC$ bounds $\overline{AC}$ from the right since $\overline{AC}$ divides $\odot ABC$ into left and right bounding arcs.

Therefore, $\odot ACD$ is covering the part of the shaded region formed by $\overline{AC}$ and $\widehat{AC}$.

D outside $\Rightarrow \angle ADC > \angle AEC \Rightarrow E$ inside

Claim. The circumcircle of each triangle in G contains no points of P in the interior.

Proof. Suppose, for contradiction that $p \in P$ is in the interior of the circumcircle of some triangle Δ in G . Let the vertices of Δ be A, B, C st. p and B falls on the opposite sides of $\overline{AC}$. Recall that G is a triangulation of P , which should make this labelling possible as p does not fall inside any $\Delta' \in G$.

Identify an arbitrary point $\pi_0 \in \text{int } \overline{AC}$ (under general position assumption)

Shoot a ray ℓ from π_0 to p , which passes a sequence of triangles $\Delta_0, \Delta_1, \dots, \Delta_t$ satisfying

(i) Δ_0 is ΔABC . (ii) Δ_i and Δ_{i+1} share an edge for each $i \in [0, t-1]$. (iii) p is a vertex of Δ_t .

Now we show the edge shared by Δ_t and Δ_{t+1} must be illegal.

Lemma 3. For each $i \in [0, t-1]$, define π_i as the point where p leaves Δ_i .
the circumcircle of Δ_i denoted as $\odot_i$ covers segment $\overline{\pi_i p}$.

全局最优
与局部最优
矛盾.

Proof. By induction.

Base Case: Follows directly from definition of π_0 and p .

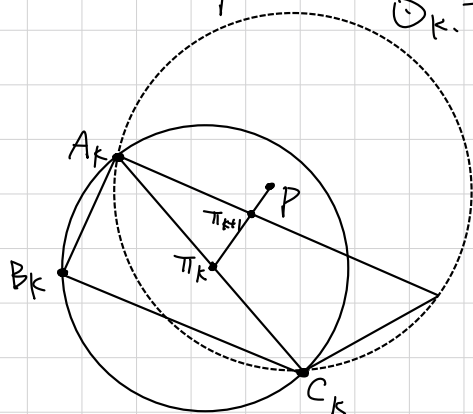
Inductive Step: Assume the lemma holds for $i \in [0, t-2]$.

For $i \geq k+1$, name vertices of Δ_k as A_k, B_k, C_k st. B_k and p falls on different sides of $\overline{A_k C_k}$.

By inductive hypothesis, $\odot_k$ covers $\overline{\pi_k p}$.

Denote D the vertex of Δ_{k+1} other than A_k, C_k . By legality, D must fall outside $\odot_k$. Then, by Lemma 2, $\odot_{k+1}$ covers $\overline{A_k C_k}$ in $\odot_k$ and hence $\overline{\pi_{k+1} p}$.

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Since $\odot_{k+1}$ covers $\overline{\pi_{k+1} p}$, p must be in the interior of $\odot_{k+1}$.

However, this contradicts the fact that the edge shared by Δ_{k+1} and Δ_k is legal.

Incremental Construction:

1. Insert d_3, d_2, d_1 to a point location structure
2. Insert $p_1, p_2, \dots, p_n$ in random order
 - a. Use the point location graph to find the triangle $p_i p_j p_k$ that contains p_r
 - b. Split the triangles into $p_r p_i p_j$, $p_r p_i p_k$, $p_r p_j p_k$.
 - c. Call $\text{legalize}(p_r, p_i p_j)$, $\text{legalize}(p_r, p_j p_k)$, $\text{legalize}(p_r, p_k p_i)$.
 - d. Update the graph
3. Remove the triangles incident on d_3, d_2, d_1 .

$\text{legalize}(p_r, p_i p_j)$

if $p_i p_j$ is illegal

let $p_j p_i p_k$ be opposite $p_i p_j p_r$

flip $p_i p_j$ to $p_r p_k$

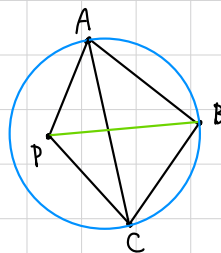
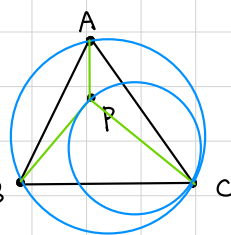
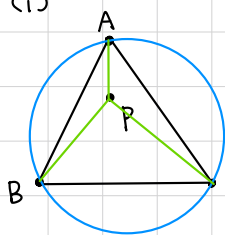
$\text{legalize}(p_r, p_i p_k)$

$\text{legalize}(p_r, p_j p_k)$

Argue: structural changes for each insertion is $O(1)$
expected rebucketing is $O(\log n)$ times

- Fact. (i) The new edges created due to the insertion of p must be in the edge of the $k+1$ points Delaunay triangulation
- (ii) The edges created due to edge-flipping must be in the edge of the $k+1$ points Delaunay triangulation

Proof. (i)



The circumcircle O_1 of $\triangle ABC$ is empty before p is inserted.

Let $O_2 \subset O_1$ be another circle with diameter $\overline{pA}$.

O_2 is empty in the interior, $\overline{pA}$ is a Delaunay edge.

Follows $\overline{pB}$, $\overline{pC}$.

Legalize flip create Delaunay edges:

Circumcircle O_1 of ABC contains p and no other point

Let $O_2 \subset O_1$ be the circle with diameter $\overline{pB}$.

O_2 is empty in the interior, $\overline{pB}$ is a Delaunay edge.

d_{-3}, d_{-2}, d_{-1} : dummy points gigantic

$p_1, p_2, \dots, p_n$: permutation of P

The cost of processing $p_i = \mathcal{O}(\text{\#edges checked}) \stackrel{\text{def}}{=} \mathcal{O}\left(\sum_{i=1}^n X_i\right)$.

Lemma: $\mathbb{E}[X_i] = 1$.

G_i = The Delaunay triangulation of $\{d_{-1}, d_{-2}, d_{-3}, p_1, p_2, \dots, p_i\}$

X_i = degree of p_i in G_i . p has to affect other triangles' circumcircle

Case: $i=n$ $\mathbb{E}[X_n] = \mathcal{O}(1)$

Case: $i < n$ $\mathbb{E}[X_i] = \sum_{\pi} \mathbb{E}[X_i | \pi] \mathbb{P}(\pi)$

$$= \frac{i!}{n!} \sum_{\pi} \mathcal{O}(1)$$

$$= \frac{i!}{n!} \cdot \frac{n!}{i!} = \mathcal{O}(1)$$

Easy implementation: Bowyer-Watson's Algorithm

- No point location structure required
- Maintain the list of vertices
- Break polygonal hole and retriangulate.

Lecture 26 - VC dimension, ϵ -samples, ϵ -nets

Definition 1. A range space S is a pair $(X, \mathcal{R})$, where X is a ground set and $\mathcal{R}$ is a family of subsets of X .

Definition 2. Let $S = (X, \mathcal{R})$ be a range space. For $Y \subseteq X$, let

$$\mathcal{R}_{|Y} = \{r \cap Y \mid r \in \mathcal{R}\}$$

denote the projection of $\mathcal{R}$ on Y . The range space S projected to Y is $S_{|Y} = (Y, \mathcal{R}_{|Y})$.

If $\mathcal{R}_{|Y}$ contains all subsets of Y , then Y is shattered by $\mathcal{R}$.

Definition 3. The Vapnik-Chervonenkis dimension of S , denoted by $\dim_{VC}(S)$, is the maximum cardinality of a shattered subset of X . If there are arbitrary large shattered subsets, then $\dim_{VC}(S) = \infty$.

Examples. ① Intervals in $\mathbb{R}$

$$\dim_{VC}(\mathcal{H}_{\text{int}}) = 2$$

② Halfplanes / Halfspaces in $\mathbb{R}^d$

$$\dim_{VC}(\mathcal{H}_{\text{half}}) = d+1$$

③ Axis-parallel rectangles in $\mathbb{R}^d$

$$\dim_{VC}(\mathcal{H}_{\text{rect}}) = 2d.$$

④ d -convex polygon in $\mathbb{R}^2$

$$\dim(\mathcal{H}_{\text{poly}}) = 2d+1.$$

Lemma (Sauer's Lemma). If $(X, \mathcal{R})$ is a range space of VC dimension δ with $|X| = n$, then $|\mathcal{R}| \leq \Phi_\delta(n)$, where

$$\Phi_\delta(n) = \sum_{i=0}^{\delta} \binom{n}{i} \leq \sum_{i=0}^{\delta} \frac{n^i}{i!} \leq n^\delta.$$

Claim. For all $n, \delta \geq 1$, we have $\Phi_\delta(n) = \Phi_\delta(n-1) + \Phi_{\delta-1}(n-1)$.

Proof. (Base Case) ^① $n=0 \Rightarrow \delta=0$, $\Phi_\delta(n)=1$.

^② $\delta=0$, $\Phi_\delta(n)=1$.

Suppose the otherwise that $|\mathcal{R}| \geq 2$. Then, let $Q_1, Q_2 \in \mathcal{R}$.

Since $Q_1 \neq Q_2$, either $Q_1 \setminus Q_2$ or $Q_2 \setminus Q_1$ is non-empty.

WLOG, suppose $e \in Q_1 \setminus Q_2$, i.e. $e \in Q_1$ and $e \notin Q_2$.

Then, $\{e\}$ can be shattered by $\mathcal{R}$. This contradicts $\delta=0$.

(Inductive Case) Suppose the lemma is true for (n_0-1, δ_0) and (n_0-1, δ_0-1) .

Then, for $(n, \delta) = (n_0, \delta_0)$.

Let e be any element of X .

Consider the sets $X' = X \setminus \{e\}$.

a partitioning of $\mathcal{R}$

$$\begin{aligned} \mathcal{R}' &= \{Q \in \mathcal{R} \mid e \in Q, Q \setminus \{e\} \in \mathcal{R}\}, \\ \mathcal{R}'' &= \{Q \in \mathcal{R} \mid e \in Q, Q \setminus \{e\} \notin \mathcal{R}\}, \\ \mathcal{R}''' &= \{Q \in \mathcal{R} \mid e \notin Q\}. \end{aligned}$$

$\mathcal{R}_1 = \{Q \setminus \{e\} \mid Q \in \mathcal{R}'\}$, $\mathcal{R}_2 = \{Q \setminus \{e\} \mid Q \in \mathcal{R}''\} \cup \mathcal{R}'''$. one-to-one correspondence from $\mathcal{R}$ to $\mathcal{R}_1 \cup \mathcal{R}_2$

① The VC-dimension of $S_1 = (X', \mathcal{R}_1)$ is at most $\delta-1$.

Suppose the otherwise that the statement is false, then

$\exists S' \subseteq X'$, s.t. ① $|S'| = \delta$, ② S' can be shattered by $\mathcal{R}_1$.

Let $S = S' \cup \{e\} \subseteq X$. Claim: S can be shattered by $\mathcal{R}$. $\leftarrow |S| = \delta+1$ but $\dim_{VC}(S) = \delta$.

Goal: $\forall T \subseteq S, \exists Q \in \mathcal{R}$ such that $T = Q \cap S$.

(a) $e \notin T \Rightarrow T$ is a subset of S'

As S' is shattered by $\mathcal{R}_1$,

$\exists Q \in \mathcal{R}_1$ such that $T = Q \cap S' = Q \cap S$.

As $Q \in \mathcal{R}$, Q serves the purpose

(b) $e \in T \Rightarrow T \setminus \{e\}$ is a subset of S'

As S' is shattered by $\mathcal{R}_1$, $\exists Q' \in \mathcal{R}_1$ such that $T \setminus \{e\} = Q' \cap S'$

$T = (Q' \cup \{e\}) \cap S = Q \cap S$.

② The VC-dimension of $S_2 = (X', \mathcal{R}_2)$ is at most δ .

Suppose the otherwise that the statement is false, then

$\exists S' \subseteq X'$, s.t. ① $|S'| = \delta+1$, ② S' can be shattered by $\mathcal{R}_2$.

Claim: $S' \subseteq X$. * S' can be shattered by $\mathcal{R}$.

Goal: $\forall T \subseteq S', \exists Q \in \mathcal{R}$ such that $T = Q \cap S'$.

$\exists Q' \in \mathcal{R}_2$ such that $T = Q' \cap S'$. ① $Q' \in \mathcal{R}''$. Note that $S' \subseteq X$ and thus $e \notin S'$. $Q' \cap S' = (Q' \cup \{e\}) \cap S'$. $Q' \cup \{e\} \in \mathcal{R}$.